Hyper-Local Aviation Weather Forecasting & Verification System

INDIVIDUAL THESIS

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ABSTRACT

Short-term aviation weather forecasting is important for safe and efficient airport operations, particularly in tropical environments where atmospheric conditions can change rapidly. The wider research project addresses hyper-local aviation weather forecasting and verification for Bandaranaike International Airport (VCBI) under Sri Lankan atmospheric conditions. This individual study focuses on the pressure–humidity–temperature component identified in the project topic assessment form and, in the implemented research, concentrates on a hybrid Random Forest (RF) and Long Short-Term Memory (LSTM) approach for short-term temperature and pressure prediction. Historical VCBI meteorological observations are preprocessed chronologically and transformed into temporal and meteorological features, including lagged variables, rolling averages and interaction features. RF is used to model nonlinear relationships among meteorological variables, while LSTM is used to capture temporal dependencies in sequential observations. Their outputs are combined to produce three-hour-ahead forecasts of temperature and pressure. The predicted values are subsequently transformed into aviation-relevant Density Altitude (DA) using pressure altitude and International Standard Atmosphere relationships. Forecast performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2), with future observations used for verification. The research paper reports an approximate R^2 of 0.84 and MAE of about 0.86 for the hybrid approach, compared with approximately 0.68 for RF and 0.25 for LSTM. The individual contribution therefore demonstrates how combining tabular nonlinear learning and sequential learning can improve short-term hyper-local temperature and pressure forecasting and translate the predictions into an aviation-oriented indicator.

Keywords: Hyper-local aviation weather, Random Forest, LSTM, pressure prediction, Density Altitude

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
DA	Density Altitude
DL	Deep Learning
ICAO	International Civil Aviation Organization
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
METAR	Meteorological Aerodrome Report
ML	Machine Learning
NWP	Numerical Weather Prediction
PA	Pressure Altitude
RH	Relative Humidity
RF	Random Forest
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
TAF	Terminal Aerodrome Forecast
VCBI	Bandaranaike International Airport

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1 INTRODUCTION

1.1 Overall project context

The weather forecast is an important component in aviation decision-making, as the atmosphere may affect decisions related to flight planning, runway operations and performance of the aircraft. The aviation industry relies on weather observations and forecasts such as METAR and TAF in order to comprehend current as well as future weather conditions. The proposed broader study is a hyper local aviation weather forecasting and validation system for Bandaranaike International Airport (VCBI) based on four Sri Lankan seasons: Southwest Monsoon, Northwest Monsoon, First Inter-Monsoon, Second Inter-Monsoon.

Localized forecasts are required since the larger numerical models used to forecast the weather conditions are made for regional or global atmospheres and may not completely model the sudden changes in the atmosphere occurring at that airport. The study thus takes an approach that is airport-based, wherein the local data and aviation-related calculations have been included in the process.

In the group research paper, the model is constructed within the context of Bandaranaike International Airport (VCBI) and the significance of short-term weather forecasting. The airport-centered approach is significant since the study does not aim to improve regional numerical weather predictions. Rather, it seeks to identify certain patterns that might be unique only to the local environment using local observations. In the group's research paper, it is indicated that temperature, pressure, humidity, wind, visibility, and clouds are some of the weather factors that can impact flight and runway operations. It is also stated in the TAF that large-scale forecasts generated by models such as the Global Forecast System and ECMWF are dependable and accurate; however, they do not account for localized atmospheric conditions at an individual terminal. Therefore, the individual component will be part of a smaller research problem.

There are several forecasting subcomponents in the whole system, but each thesis should differentiate the contribution of Sankadeepa W.D.I from the other members of the team. In the TAF, the given subcomponent is assigned to be responsible for coupled prediction of pressure-humidity-temperature for the Northeast Monsoon. In the present research paper, the subsystem which was developed by implementing the temperature and pressure is defined as a combination of RF and LSTM. RF can be used to represent the nonlinear relationship of the meteorological factors, and the LSTM is responsible for the sequential relationships. Two outputs will be combined for forecasting temperature and pressure for three hours ahead.

Temperature and pressure have not been considered as separate numerical forecast target variables in the implemented module. In the research paper, the numerical predictions have been converted into Pressure Altitude, International Standard Atmosphere Temperature, deviation from ISA Temperature and Density Altitude. This is important since the forecast will be helpful for an aviation user only when the prediction can be interpreted in terms of aircraft flight conditions. Using the formulas provided in the research paper, the predicted pressure and temperature have been converted to an aviation-relevant indicator called "Density Altitude" and classified as either Normal, Elevated or Critical based on the criteria used in the system.

1.2 Individual component context

The original topic assessment form described the assigned task as a pressure–humidity–temperature coupled prediction model and referred to the Northeast Monsoon as its initial context. The implemented individual model is discussed here as a short-term VCBI forecasting component using the available historical observations, rather than as a model restricted to one seasonal period.

During implementation, the approach was refined into a hybrid RF–LSTM forecasting subsystem. The research paper describes RF as the component that captures nonlinear relationships among meteorological variables and LSTM as the component that captures sequential dependencies. The two outputs are combined for three-hour temperature and pressure forecasting. This implemented approach is the primary technical focus of this individual thesis.

The Topic Assessment Form originally framed the individual task as a pressure–humidity–temperature coupled prediction model. Its seasonal wording is recorded as part of the project history; the thesis does not claim that the implemented model is limited to one monsoon period. The listed tasks include collecting historical VCBI weather data, preprocessing time-series observations, building a multi-variable machine-learning model, evaluating prediction accuracy and forecast verification, and creating weather visualizations. The novelty statement in the TAF is that the model should learn relationships among pressure, humidity and temperature rather than treating each variable as completely independent. This original specification establishes the individual research direction and explains why interactions among meteorological variables are included in feature engineering.

The group research paper reports a more concrete implementation than the original TAF wording. The temperature and pressure subsystem was implemented as a hybrid RF–LSTM approach. The RF model captures nonlinear relationships among humidity, dew point, wind and lagged variables, while the LSTM processes sequences of meteorological observations. The reported results distinguish the individual RF baseline, the LSTM baseline and the combined hybrid approach. The paper reports approximately 68% R^2 for RF, approximately 25% for LSTM and approximately 84% for the hybrid approach, with an MAE of approximately 0.866 for the hybrid. These values are retained exactly as reported at the level of precision available in the source; the thesis does not invent additional decimal places or unreported hyperparameters.

The individual work is deliberately narrower than the complete system. The other modules include wind prediction using XGBoost and Temporal Fusion Transformer, cloud and visibility prediction using XGBoost, and a separate pressure/dewpoint/RH subsystem using LightGBM. Those components are discussed only when explaining integration and system context. The main experimental focus of this thesis is the temperature and pressure forecasting component, its time-series preparation, feature engineering, RF and LSTM modelling, hybrid combination, aviation transformation and verification. This separation prevents the individual thesis from claiming the work of the other members as the candidate’s own contribution.

1.3 Problem statement

Airport weather can change over short time periods, and accurate localized temperature and pressure information is important for aviation operations. General weather prediction research in Sri Lanka has demonstrated the usefulness of machine learning for weather and monsoon prediction, but the reviewed work in the project paper does not directly address joint temperature, atmospheric pressure and relative-humidity-related prediction at VCBI under Sri Lankan atmospheric conditions. A further challenge is that meteorological time series contain both nonlinear relationships among variables and temporal dependencies. A single modelling approach may not represent both characteristics equally well. Therefore, this research investigates a hybrid RF–LSTM approach for short-term temperature and pressure prediction and translates the predictions into an aviation-relevant Density Altitude indicator.

The project problem is not simply that weather is difficult to predict. The more specific problem is that airport operations require timely local information while many conventional forecasting systems operate at larger spatial scales. The TAF identifies localized atmospheric variation as a limitation when regional or global models are applied directly to an individual terminal. Sri Lanka adds another source of complexity because the atmospheric regime changes through the Southwest Monsoon, Northeast Monsoon, First Inter-Monsoon and Second Inter-Monsoon periods. A local model therefore has to deal with temporal dependence, interactions among variables and changing seasonal conditions.

Temperature, pressure and humidity are physically and statistically related. The original individual component was therefore framed as a coupled prediction problem rather than three unrelated regression tasks. The implemented research paper does not report a single multi-output neural architecture for all three variables; instead, it reports the RF–LSTM pressure subsystem and a separate LightGBM dewpoint/pressure/RH subsystem elsewhere in the complete project. The individual thesis records this distinction explicitly. It is an important methodological point because the final implementation evolved from the original component concept into a modular set of models connected through shared meteorological features and physical relationships.

Forecasting alone does not establish reliability. The project therefore stores forecasts together with their target timestamps and later matches them against future METAR observations. The verification stage calculates regression metrics including MAE, RMSE and R^2. The process is chronological: a prediction made for a future time is compared with the observation that actually becomes available at that target time. This structure prevents the thesis from treating a model's training performance as equivalent to operational forecasting performance. It also creates a basis for later monitoring and model maintenance because forecast errors can be stored and summarized over time.

1.4 Research gap

The research gap is derived from the reviewed literature and the project paper. Earlier Sri Lankan studies largely focus on general weather or monsoon rainfall prediction, while aviation studies address weather hazards and automated aviation forecasting. Deep-learning studies demonstrate the value of temporal or spatiotemporal modelling, while data-assimilation studies emphasize the importance of appropriate meteorological observations. However, the reviewed works do not directly address joint temperature, atmospheric pressure and relative-humidity-related prediction at VCBI under Sri Lankan atmospheric conditions. The individual component addresses this gap by combining nonlinear tabular learning and sequence learning within a hyper-local airport forecasting context.

The literature summarized in the research paper includes work on monsoon rainfall prediction using artificial neural networks and short-term weather prediction using machine learning in Sri Lanka. These studies demonstrate that data-driven forecasting is feasible in the local meteorological context. However, the project paper argues that they are generally concerned with weather forecasting at a broader level rather than hyper-local airport prediction. The gap is therefore not the absence of machine learning in Sri Lankan weather research; it is the limited direct treatment of airport-specific, short-horizon forecasting under changing seasonal conditions.

Aviation-focused literature discussed in the paper addresses hazards, high-impact weather, automated TAF generation and aviation-related prediction. These studies demonstrate the operational importance of accurate weather information. However,

the project identifies a missing combination: airport-level local observations, monsoon-aware modelling, short-term forecasting and verification against aviation observations. The individual component narrows this combination further by studying temperature and pressure and by transforming the predictions into Density Altitude. This provides a concrete bridge between a machine-learning forecast and an aviation-relevant operational indicator.

The individual research gap can therefore be stated as follows: there is a need to evaluate whether a hybrid model that combines nonlinear tabular learning with sequential temporal learning can provide useful short-term temperature and pressure forecasts from airport-level observations at VCBI, while preserving a clear verification pathway to future METAR observations and an aviation-oriented interpretation. This formulation remains consistent with the TAF novelty statement while reflecting the actual RF–LSTM implementation described in the group paper.

1.5 Aim and objectives

Aim

To develop and evaluate a hybrid machine-learning approach for short-term hyper-local temperature and pressure prediction at Bandaranaike International Airport, using the available historical airport observations and aviation-relevant interpretation.

Individual objectives

1. Collect and prepare historical VCBI meteorological observations for time-series modelling.
2. Engineer meteorological and temporal features that represent recent atmospheric behaviour and variable interactions.
3. Develop an RF model for nonlinear relationships among meteorological variables.
4. Develop an LSTM model for sequential temporal dependencies.
5. Combine RF and LSTM outputs to produce three-hour-ahead temperature and pressure forecasts.

6. Evaluate the forecasting performance using MAE, RMSE and R^2 and verify predictions against future observations.
7. Transform predicted temperature and pressure into Pressure Altitude, ISA temperature, ISA deviation and Density Altitude for aviation interpretation.

Each individual objective can be mapped to a concrete stage in the implemented workflow. Historical VCBI observations are used for data preparation; lagged variables, rolling statistics and meteorological interactions are used for feature engineering; RF provides a nonlinear tabular baseline; LSTM models temporal sequences; the hybrid combines their predictive outputs; MAE, RMSE and R^2 are used for evaluation; and the predicted temperature and pressure values are converted into aviation metrics. This mapping is useful for assessment because it makes clear that the thesis is not merely describing an AI model. It follows a complete research-development chain from data to interpretation and verification.

The main objective is considered achieved at the level supported by the current paper when the implementation can generate the specified short-term forecasts and the reported results can be evaluated against the individual baselines. The research paper provides the main quantitative evidence: RF approximately 68% R^2, LSTM approximately 25% R^2 and hybrid approximately 84% R^2, with hybrid MAE approximately 0.866. The paper also demonstrates downstream Density Altitude calculation. Exact final test-set distributions, confidence intervals and per-period validation statistics are not reported in the source and are therefore not fabricated in this thesis.

Table 1.1.Mapping of Individual Objectives to the Implemented Workflow

Individual Objective	Implementation / Evidence
Collect and prepare historical VCBI meteorological observations.	Historical airport weather observations are prepared for time-series modelling.
Engineer meteorological and temporal features.	Lagged variables, rolling statistics and meteorological interaction features are generated.
Develop a Random Forest (RF) model.	RF is used to model nonlinear relationships among meteorological variables.
Develop an LSTM model.	LSTM is used to learn sequential temporal dependencies in weather observations.
Combine RF and LSTM outputs for three-hour-ahead forecasts.	The hybrid approach combines predictive outputs to forecast temperature and pressure at T+3.
Evaluate and verify forecasting performance.	MAE, RMSE and R^2 are used for performance evaluation, with predictions compared against future observations.
Derive aviation-relevant metrics from predicted weather variables.	Predicted temperature and pressure are used to calculate Pressure Altitude, ISA temperature, ISA deviation and Density Altitude.

1.6 Research questions or hypotheses

- RQ1: Can a hybrid RF–LSTM model improve short-term temperature and pressure prediction compared with the individual RF and LSTM approaches?
- RQ2: To what extent do temporal and meteorological interaction features contribute to short-term hyper-local forecasting at VCBI?
- RQ3: Can predicted temperature and pressure be transformed into a meaningful aviation-relevant Density Altitude indicator?
- RQ4: How reliably can the individual component be verified against future METAR observations?

Working hypothesis: The hybrid RF–LSTM approach will provide better short-term forecasting performance than either RF or LSTM used independently because it combines nonlinear feature relationships with sequential temporal learning.

RQ1 asks whether the hybrid RF–LSTM model improves short-term temperature and pressure prediction compared with the individual RF and LSTM approaches. The source evidence directly supports a positive comparison at the reported aggregate level: the hybrid R^2 is approximately 84%, compared with approximately 68% for RF and approximately 25% for LSTM. The interpretation should remain cautious because the paper does not provide full uncertainty estimates or a complete statistical significance analysis. The result demonstrates a reported performance difference, not a universal claim that RF–LSTM will outperform all alternative architectures.

RQ2 concerns temporal and meteorological interaction features; the methodology documents lagged variables, six-hour LSTM windows, cyclic time features, rolling averages, pressure momentum and interactions such as temperature–RH and pressure–wind. However, the paper does not provide a complete ablation table showing the effect of each feature family individually. RQ3 is supported by the implemented Density Altitude transformation. RQ4 is supported by the verification design, but a complete long-term verification table is not present in the supplied paper. These boundaries are stated explicitly rather than filled with invented results.

1.7 Scope, limitations and assumptions

- Geographical scope: Bandaranaike International Airport (VCBI).
- Forecast horizon: three hours ahead for the individual temperature and pressure subsystem.
- Primary prediction variables: temperature and pressure, with humidity-related and other meteorological variables used as features where available.
- The wider system uses multiple weather modules; this thesis focuses on the individual RF–LSTM pressure component.
- Historical VCBI data described in the paper span 2019–2026 and contain more than 120,000 samples.

- Training and testing are separated chronologically without shuffling to preserve the time-series nature of the problem.
- Reported results are based on the current research paper and should be replaced or expanded with the final experimental logs and complete test-set statistics before final submission.
- The research does not involve human participants or personal data.

The historical dataset is described as VCBI meteorological data from 2019–2026 with more than 120,000 samples. Current METAR observations are obtained from the Aviation Weather Center and stored in SQLite for the wider workflow. The individual study therefore has a clearly defined geographic boundary at VCBI. Supporting satellite or global meteorological data are mentioned in the TAF as possible additional inputs, but the current paper's individual implementation description does not establish that all such sources were actually used in the RF–LSTM model. They are consequently treated as possible supporting data rather than as confirmed model inputs.

The primary individual forecast horizon is T+3 hours. The wider system also includes a T+1 wind forecast, but that is outside the individual experimental scope. Chronological splitting without shuffling is a central assumption because random shuffling could allow future patterns to influence the training set and weaken the validity of a time-series evaluation. The model is also interpreted as a decision-support component rather than an autonomous flight-safety authority. The project paper describes pilot-oriented guidance and confidence information, not automated operational control of aircraft.

1.8 Individual contribution

The individual contribution is the design, implementation and evaluation of the hybrid RF–LSTM temperature and pressure forecasting component. This includes time-series data preparation, feature engineering, model development, combination of model outputs, evaluation, forecast verification and conversion of predictions into Density Altitude. The individual work is integrated into the wider group system, but the

detailed research contribution remains the modelling and aviation interpretation of temperature and pressure.

The individual research contribution is the investigation and implementation of a hybrid data-driven approach for short-term airport-level temperature and pressure forecasting. The contribution combines three levels: meteorological time-series preparation, machine-learning modelling, and aviation-oriented interpretation. The RF and LSTM models are not treated only as algorithm names; their roles are differentiated according to the type of pattern they are intended to capture. The hybrid result is then evaluated against the reported baselines and transformed into Density Altitude so that the model output has a clear domain interpretation.

The implementation contribution includes the preparation of the historical data, engineering of temporal and meteorological interaction features, creation of the RF and LSTM forecasting stages, combination of their outputs, and connection to the broader verification and aviation-calculation pipeline. The group paper reports FastAPI, Supabase, SQLite and React integration for the overall system. Those technologies are discussed here only to explain how the individual forecasting component is consumed by the complete solution.

1.9 Thesis structure

Chapter 1 introduces the project, individual component, problem, gap, objectives, research questions, scope and contribution. Chapter 2 reviews theoretical foundations and related research. Chapter 3 presents the methodology, data, preprocessing, modelling procedure and evaluation methods. Chapter 4 describes requirements and the design of the individual solution. Chapter 5 discusses implementation, integration, testing and operation. Chapter 6 presents research and technical results. Chapter 7 interprets the findings and discusses contribution and limitations. Chapter 8 considers commercialization and sustainability. Chapter 9 concludes the research and identifies future directions.

The thesis follows the sequence required by the IT4010 individual thesis template. Chapter 1 establishes the problem and individual contribution. Chapter 2 explains the scientific and technical background and positions the work against related studies.

Chapter 3 describes the research methodology and evaluation design. Chapter 4 translates the research requirements into a solution design. Chapter 5 documents implementation, integration and testing. Chapter 6 presents the results that are actually supported by the research paper. Chapter 7 interprets those results and identifies validity threats. Chapter 8 considers value, sustainability and future adoption. Chapter 9 concludes the work and identifies future research directions. The appendices provide detailed evidence artefacts without replacing the explanations in the main chapters.

The practical value of an airport-level forecasting component depends on more than the ability to produce a numerical estimate. The forecast must be linked to a clearly defined target time, use information that would genuinely have been available at the forecast origin, and be presented in a form that can be compared with a later observation. These requirements influence the design of the individual component from data preparation through verification. A three-hour horizon provides a specific operational question: given the observations available now, how closely can the system estimate temperature and pressure three hours later? Defining the question in this way makes the evaluation repeatable and prevents a model from being assessed against an ambiguous notion of a short-term forecast.

The project is positioned as decision support rather than as a replacement for official meteorological services. Aviation weather information is produced through established observation, forecasting and dissemination practices, and an academic machine-learning component should be assessed alongside those practices rather than in isolation. The model can contribute additional local estimates and an auditable record of prediction error, but a favourable research metric alone does not establish that the output is suitable for operational use. This distinction is important throughout the thesis: the work evaluates a research prototype and its technical contribution, while operational acceptance would require further independent validation, monitoring and approval by relevant aviation and meteorological authorities.

A central design consideration is the difference between predicting a variable and interpreting its consequences. Temperature and pressure are the model's primary numerical targets, whereas Density Altitude is a derived quantity that helps explain the combined effect of atmospheric conditions in an aviation context. The derived value

is not an independent observation and should not be presented as one. Its reliability depends on both the quality of the predicted inputs and the correctness of the transformation applied to them. The study therefore treats model evaluation and physical interpretation as related but distinct stages. This separation makes it possible to identify whether an error originates in the forecast itself or in the subsequent calculation and presentation.

The thesis follows an evidence-led approach. Statements about implementation are limited to the methods described in the available project artefacts, and reported metrics are retained at the precision available in the source rather than being recalculated or embellished without the underlying predictions. Where the available material does not contain a complete variable-level result, confidence interval or ablation experiment, that absence is identified as a limitation. This is preferable to presenting unsupported detail as fact. It also gives the work a clear path for improvement: each limitation can be converted into a measurable experiment or documentation task in a later research iteration.

A further consideration is the distinction between temporal resolution and forecast horizon. The fact that observations are available at a particular reporting interval does not automatically mean that the model can accurately predict every future interval. Forecasting three hours ahead requires the input window to be aligned with the target time and the verification method to account for any irregular observation schedule. If a METAR report is not issued exactly at the target time, the study must apply a transparent matching rule rather than silently selecting the nearest record. The chosen rule affects the apparent difficulty of the task and should therefore be treated as part of the research method, not as a minor implementation detail.

The scope of the thesis is deliberately constrained so that the contribution can be evaluated clearly. It does not attempt to create a complete numerical weather prediction system, certify an aviation product or replace human expertise. Instead, it studies one forecasting component within a broader project and examines how its outputs can be verified and interpreted. This narrow scope supports a more defensible account of individual work and provides a practical basis for future collaboration. The wider system can be acknowledged where integration is relevant, while the algorithms,

results and limitations discussed in detail remain focused on the temperature and pressure component.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review was structured around four themes: machine learning for weather forecasting, aviation weather forecasting, machine learning in aviation-related prediction, and aviation weather hazards and research gaps. The review sources reported in the group research paper were used as the core evidence base. The review was then narrowed to the individual problem of short-term temperature and pressure prediction under local monsoon-influenced conditions.

The literature review follows the thematic organization already established in the group paper. The first theme is machine learning for weather forecasting, covering artificial neural networks, short-term weather prediction, ConvLSTM and machine-learning-based global forecasting. The second theme is aviation weather forecasting, including high-impact hazards and automated TAF generation. The third theme is machine learning in aviation-related prediction, including Random Forest for flight-delay prediction. The fourth theme is aviation weather hazards and airport-specific climatology. Organizing the literature in this way allows the thesis to move from general forecasting capability to the specific airport and aviation problem.

A source is considered directly relevant when it addresses one or more of the following: weather prediction from historical observations, temporal or sequential machine learning, aviation weather, airport-specific hazards, or forecast post-processing and verification. The review does not treat every weather AI publication as evidence that an airport-specific model is effective. Instead, each source is used to establish a particular part of the research argument. For example, ConvLSTM supports the importance of temporal modelling, while the aviation literature supports the operational importance of accurate local weather information.

2.2 Domain and theoretical foundations

Short-term meteorological forecasting is a time-series prediction problem because observations are ordered in time and future atmospheric states depend partly on recent states. Meteorological variables are also physically related. Temperature, humidity, pressure and dew point can exhibit correlated or nonlinear behaviour, meaning that feature engineering and model selection are important.

Random Forest is an ensemble of decision trees that can represent nonlinear relationships between input features and a continuous target. It is suitable for tabular meteorological data containing interactions and lagged variables. LSTM is a recurrent neural-network architecture designed to learn dependencies across sequences. In this study, the complementary roles of RF and LSTM motivate the hybrid design: RF represents nonlinear feature relationships, while LSTM represents temporal sequence dependencies.

The model predicts pressure as defined in the individual component. Where the implementation uses a pressure value suitable for aviation calculations, that value and predicted temperature can be used to derive pressure altitude, ISA temperature, ISA deviation and Density Altitude. The precise calculation must follow the units and pressure definition used by the source implementation. This transformation provides a bridge between a numerical weather forecast and an operationally interpretable aviation parameter.

The individual component is grounded in the idea that meteorological variables do not evolve independently. Temperature, humidity and dew point have direct thermodynamic relationships, while pressure and wind provide information about atmospheric state and movement. The feature-engineering strategy in the paper reflects this by including lagged meteorological variables, dew-point depression, temperature–RH interaction, wind-speed–RH interaction, pressure–wind interaction and RH squared. These features do not prove causality, but they provide the learning algorithms with representations of relationships that may be useful for short-term prediction.

A three-hour forecast is a temporal prediction problem because the model must use information available before the forecast target time. The LSTM component is particularly suited to this structure because it processes sequences rather than treating each observation as an independent row. The paper uses six-hour LSTM windows. In contrast, the RF model operates on engineered tabular features such as lags and rolling statistics. The hybrid design therefore combines an explicit sequence representation with a feature-based representation of recent atmospheric conditions.

Random Forest is used in the research paper to model nonlinear relationships among the meteorological variables and lagged features. In a Random Forest, multiple decision-tree estimators can represent different partitions of the feature space and their predictions are aggregated. For this thesis, the important point is not an unreported implementation detail such as the exact number of trees; those parameters are not given in the source. The methodological role of RF is clear: it provides a strong tabular baseline and a complementary view of the problem to the sequential LSTM.

Long Short-Term Memory networks are recurrent neural-network architectures designed to process ordered sequences while retaining information across time steps. In the individual system, the LSTM receives a six-hour sequence representation and learns temporal dependencies in the meteorological observations. The research paper reports a lower standalone R^2 for the LSTM baseline than for RF. This does not mean that LSTM is inherently unsuitable for weather forecasting; rather, it indicates that the reported implementation and data representation performed less strongly than the RF and hybrid alternatives in the tested setting.

The hybrid approach is motivated by complementarity. RF can learn nonlinear relationships in engineered tabular variables, while LSTM can learn sequential dependencies from a time window. Combining their outputs can therefore provide a prediction that reflects both representations. The reported hybrid R^2 of approximately 84% is higher than the reported RF and LSTM baselines. The result is consistent with the design rationale, although the source does not provide a formal ablation study or uncertainty analysis that would prove exactly which component contributes how much to the improvement.

The forecasting stage becomes aviation-oriented through deterministic transformations. The paper computes Pressure Altitude using the difference between standard sea-level pressure and predicted pressure. It then computes ISA temperature using the standard sea-level temperature and the stated lapse rate, calculates ISA deviation from predicted temperature, and derives Density Altitude using the reported coefficient. This sequence separates machine-learning inference from physical interpretation. The formulas are therefore transparent and reproducible even though the underlying weather prediction is learned from data.

2.3 Related research

The paper cites Nagamulla, Rathnayake and Ratnaweera on artificial neural networks for monsoon rainfall forecasting in Sri Lanka and Hennayake et al. on machine-learning-based short-term weather prediction. These studies are important because they establish local evidence that historical meteorological information can support predictive modelling. The individual thesis extends the problem context from general weather or rainfall prediction to airport-specific, short-horizon aviation forecasting. The extension is therefore geographic and operational as well as algorithmic.

Shi et al.'s ConvLSTM work is used in the paper to illustrate how deep-learning models can capture temporal and spatial structure in weather-related data. The individual system is not a ConvLSTM implementation; it uses an LSTM sequence model alongside RF. The relevance is methodological: weather forecasting requires models that can represent time-dependent behaviour. The use of a six-hour LSTM window in the individual component follows this broader principle while keeping the implementation focused on local airport observations.

The aviation literature in the paper includes high-impact weather reviews and AI-assisted TAF generation. These studies reinforce that aviation forecasting has operational requirements beyond generic weather prediction. A forecast for an airport must be timely, interpretable and connected to aviation-relevant variables.

2.4 Existing systems, products and techniques

Operational aviation weather relies on products such as METAR and TAF. Large-scale numerical weather prediction models such as the Global Forecast System and ECMWF provide regional and global forecasts. The research project does not attempt to replace such systems at the global scale. Instead, it provides a hyper-local airport-level forecasting layer based on local observations and AI models. The individual component complements this architecture by focusing on short-horizon temperature and pressure prediction.

From a modelling perspective, the paper evaluates or reports several algorithms across different components. For the individual pressure component, RF and LSTM form the hybrid approach. The wind module uses XGBoost and Temporal Fusion Transformer, the cloud and visibility module selects XGBoost, and another pressure/dewpoint subsystem uses LightGBM with a physical humidity calculation. These other components are included here only to establish the system context; they are not claimed as the individual research contribution.

METAR provides an observation of airport weather conditions, while TAF provides a forecast product for an aerodrome. The project uses these products differently: current METAR is part of the real-time data pipeline and future METAR observations are used as ground truth for verification. The project also compares its generated forecasts with official aviation information in the wider system. This distinction between prediction input, target observation and comparison forecast is essential for a rigorous verification architecture.

The TAF identifies GFS and ECMWF as examples of large-scale numerical weather prediction systems. Their strength is regional and global atmospheric representation, but the project motivates local machine learning because airport observations can contain site-specific patterns. The proposed approach does not claim to replace NWP. Instead, it represents an additional hyper-local forecasting layer based on local historical observations. This positioning is more defensible because it recognizes the complementary roles of large-scale atmospheric models and local data-driven methods.

The project paper references neural networks, ConvLSTM, Random Forest, automated TAF systems and other machine-learning approaches. The individual implementation selects RF and LSTM because their strengths are complementary to the available local time-series representation. The selection is therefore driven by the problem structure described in the paper rather than by a claim that one algorithm is universally best. The reported results provide an empirical basis for retaining the hybrid approach in the current system.

2.5 Gap synthesis and conceptual framework

The synthesized gap is the lack of a directly targeted hyper-local approach for joint temperature and pressure-related prediction at VCBI under Sri Lankan atmospheric conditions. The conceptual framework of the individual research is: historical local observations → preprocessing and feature engineering → RF nonlinear learning + LSTM sequence learning → hybrid pressure forecasts → forecast verification → aviation transformation to Density Altitude.

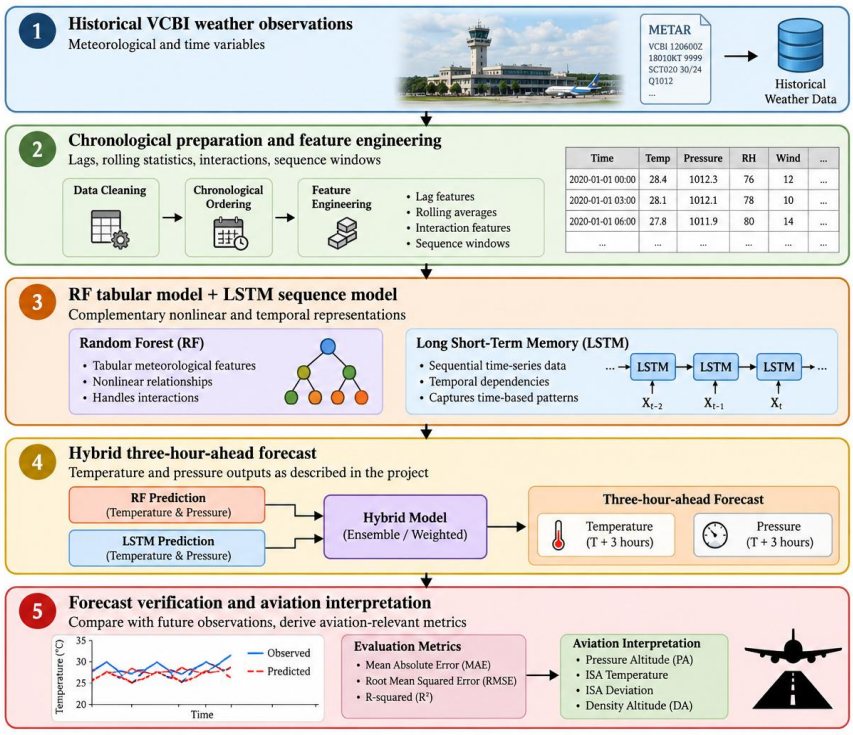


Figure 2.1.Individual pressure and temperature forecasting framework

The individual conceptual framework can be represented as five connected stages: local historical observations; preprocessing and feature engineering; complementary RF and LSTM learning; hybrid T+3 prediction; and aviation interpretation plus verification. Historical observations provide the evidence base. Feature engineering represents recent temporal behaviour and meteorological interactions. RF and LSTM learn different aspects of that representation. The hybrid output is then converted into Pressure Altitude, ISA temperature, ISA deviation and Density Altitude. Finally, the forecast is verified against future METAR observations. This framework links the research gap directly to an implementable system component.

The contribution is not simply the use of a familiar algorithm. Its value lies in applying a hybrid modelling strategy to a hyper-local aviation context at VCBI, using local historical observations, a short forecast horizon and a transparent aviation transformation. The paper's reported hybrid R^2 of approximately 84% provides initial evidence that the combination is promising. The contribution should nevertheless be interpreted as a VCBI-focused research and engineering result rather than a general proof of performance across all airports or all monsoon regimes.

The literature should be interpreted according to the spatial scale, forecast horizon, target variables and intended use of each study. A rainfall model evaluated over a district, for example, does not provide a directly comparable baseline for a three-hour temperature forecast at a single airport. Likewise, a global numerical weather prediction system and a station-level machine-learning model operate with different input information and design objectives. Their results can inform the rationale for this research, but a numerical comparison is meaningful only when the evaluation datasets, targets, units and verification periods are sufficiently aligned. This thesis therefore uses related research to establish context and identify methodological opportunities rather than to claim superiority over systems that were not tested on the same data.

Random Forest and LSTM represent different modelling assumptions. Random Forest combines predictions from many decision trees and can model nonlinear relationships among tabular predictors without requiring a linear response form. It is useful where interactions among recent weather observations and engineered features are informative. LSTM is a recurrent neural-network architecture designed to retain

information across a sequence through gated state updates. It can learn temporal dependencies from ordered observations, but its performance depends on sequence construction, scaling, training choices and the amount and quality of available data. The rationale for a hybrid approach is therefore complementary representation, not an assumption that combining models must always improve performance.

A fair assessment of a hybrid model requires the component models to be evaluated under comparable conditions. If the Random Forest model uses one test period and the LSTM model uses a different period, the observed difference may reflect weather difficulty rather than model quality. The same concern applies when one model is evaluated on more records because of different preprocessing rules. A robust comparison uses consistent target timestamps, comparable input availability and an identical held-out verification interval wherever possible. The reported project paper provides aggregate comparison values, but the available evidence does not include every detail needed to independently reconstruct all comparisons. The thesis consequently reports the supplied values cautiously and identifies common evaluation conditions as an area for stronger reproducibility.

The review also distinguishes statistical forecasting from aviation interpretation. A low error for temperature or pressure is not by itself proof that a derived Density Altitude category is reliable, because errors can propagate through the transformation and category thresholds may amplify small numerical differences near a boundary. Related aviation-weather work therefore motivates evaluating both continuous prediction errors and the behaviour of the derived indicator. For a complete future evaluation, the derived quantity should be calculated from forecast inputs and from the corresponding observed inputs using the same documented equations. Differences can then be assessed directly, with particular attention to cases where a small error changes the displayed operational category.

Forecasting studies also differ in how they handle the persistence of weather variables. A simple persistence forecast assumes that the next value will remain close to the latest observation. Although such a baseline may be limited, it is valuable because many atmospheric variables change gradually over short intervals. A machine-learning model should ideally be compared with persistence as well as with

more complex algorithms, since beating a sophisticated model is not the only meaningful benchmark. The available project results report RF, LSTM and hybrid comparisons, but do not establish a complete persistence-baseline analysis. Adding that baseline in future work would clarify the incremental value of model complexity over a straightforward reference forecast.

Another theme in the literature is the difference between predictive performance and interpretability. Tree-based models can provide feature-importance summaries, although such summaries do not necessarily establish causation and may be affected by correlated predictors. Recurrent neural networks can learn sequential patterns but are generally more difficult to explain directly. A hybrid system therefore benefits from clear documentation of its inputs, forecast horizon, transformations and evaluation process. In this research, interpretability is also supported by the downstream Density Altitude calculation, which connects the outputs to a defined physical relationship. This form of interpretability complements, but does not replace, analysis of the model's internal behaviour.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows an experimental, quantitative machine-learning research design. Historical meteorological observations are transformed into supervised time-series learning examples, models are trained on earlier observations and evaluated on later observations. Chronological splitting is used rather than random shuffling to reduce temporal leakage and to better represent future forecasting.

The individual work follows an empirical predictive-modelling design. Historical VCBI observations are transformed into supervised time-series examples, models are trained using earlier observations and evaluated using later observations. The chronological split is important because the practical use case is forward forecasting. The design therefore avoids random shuffling, which could allow observations from later periods to influence training. The paper reports the use of this chronological

approach for the main dataset and an 80/20 non-shuffled split for the separate pressure/dewpoint subsystem.

The individual experiment compares three modelling conditions: RF alone, LSTM alone and the hybrid combination. This is useful because the research question concerns whether combining the two representations produces a better forecast. The comparison is based on common forecasting metrics, principally R^2 and MAE as reported in the paper, with RMSE also defined as part of the verification methodology. The result is therefore a model comparison rather than a single-model demonstration.

3.2 Data, participants or study materials

No human participants are involved. The research uses meteorological and aviation weather data. The paper reports historical VCBI meteorological data from 2019–2026 with more than 120,000 samples and current METAR observations obtained from the Aviation Weather Center and stored in an SQLite database. The data contain atmospheric variables including temperature, wind speed, wind direction, pressure, humidity and related measurements.

The research paper states that historical VCBI meteorological data from 2019 to 2026 contain more than 120,000 samples. The variables discussed include temperature, wind speed and direction, pressure, humidity and dew point. The data are used as the historical foundation for training the forecasting models. The thesis does not claim an exact final row count because the source gives only “over 120,000” rather than a precise value. This is an important example of preserving source accuracy rather than inventing precision.

Current METAR observations are obtained from the Aviation Weather Center and stored in an SQLite database as part of the wider system. The paper describes a pipeline that retrieves, parses and stores relevant observations before prediction and later verification. This creates a consistent structure between training data and operational inference. The source does not provide a complete database schema or every API field, so this thesis describes the confirmed variables and data flow without fabricating an undocumented schema.

The TAF explicitly states that the research does not involve human participants, personal data or sensitive information. The study relies on meteorological and environmental datasets. Therefore, the individual component does not require a human-subject survey or interview as part of the reported research. Any future stakeholder evaluation would be a separate validation activity and should not be represented as completed unless evidence is available.

Table 3.1 Key meteorological variables and their roles

Variable / feature family	Role in individual model
Temperature	Target for three-hour-ahead temperature forecasting
pressure / pressure	Target for three-hour-ahead pressure forecasting and aviation calculations
Relative humidity	Predictive meteorological feature / interaction context
Dew point	Predictive feature and atmospheric relationship context
Wind variables	Predictive context through lagged and interaction features
Lagged variables	Represent recent atmospheric state
Rolling averages	Represent local temporal trends
Cyclic time features	Represent daily/seasonal periodicity
Pressure momentum / pressure movement	Represent recent pressure change

3.3 Experimental or study procedure

- Acquire and consolidate historical VCBI observations and current METAR observations.
- Order observations chronologically and remove duplicates.
- Handle missing values according to the implemented preprocessing workflow.

- Engineer temporal and meteorological features including lags, rolling averages and interaction terms.
- Create sequential windows for LSTM modelling; the paper reports six-hour LSTM windows in the feature engineering process.
- Split training and testing data chronologically without shuffling.
- Train the Random Forest model for nonlinear tabular relationships.
- Train the LSTM model for sequential dependencies.
- Combine the RF and LSTM outputs to generate three-hour-ahead temperature and pressure predictions.
- Evaluate forecasts with MAE, RMSE and R^2.
- When the future observation becomes available, match the forecast timestamp to the corresponding METAR observation and calculate the forecast error.
- Convert predicted pressure and temperature into Pressure Altitude, ISA temperature, ISA deviation and Density Altitude.

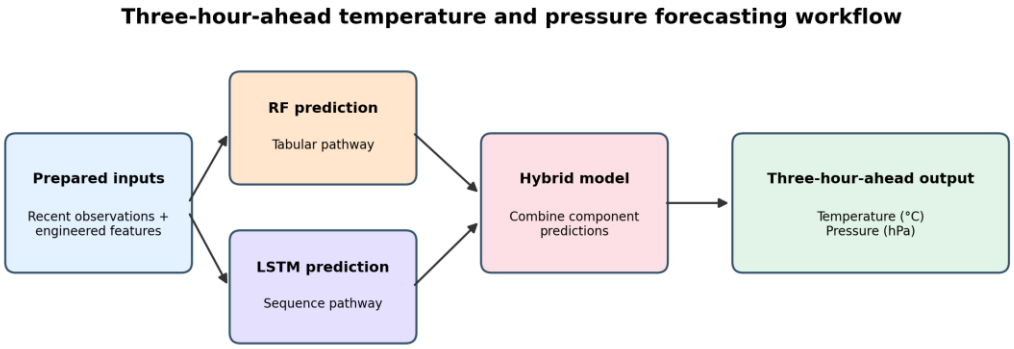


Figure 3.1.Hybrid RF–LSTM prediction architecture

The reported preprocessing sequence begins with chronological ordering and deduplication, followed by treatment of missing values and creation of temporal features. Where required, Min-Max normalization is applied. The data are then transformed into features representing recent atmospheric behaviour. This sequence is

important because preprocessing is part of the predictive experiment: a model cannot learn reliable relationships if duplicate or temporally inconsistent observations remain in the training data.

The paper lists several engineered features: lagged meteorological variables, six-hour LSTM windows, sine-cosine encoding of wind direction, dew-point depression, temperature–RH interaction, wind-speed–RH interaction, pressure–wind interaction, RH squared, cyclic time features, pressure momentum and three- and six-hour rolling averages. These features serve different purposes. Lags and rolling averages describe recent state; cyclic features represent periodic time; interaction terms represent relationships among variables; and sequence windows provide the LSTM with ordered context.

After feature preparation, the RF and LSTM models are trained using the chronological training portion. The RF receives engineered tabular representations, while the LSTM receives sequential windows. The model outputs are combined to obtain the T+3 forecast. The source does not specify the exact mathematical weighting or stacking equation used to combine the outputs. Therefore, the thesis describes the combination as a hybrid output without inventing an undocumented weight or meta-model.

Each generated forecast is stored with its target timestamp. When the corresponding future observation becomes available, the verification stage matches the prediction to the observed value. The paper states that MAE, RMSE and R^2 are used for regression verification. This makes the evaluation operationally meaningful because the prediction is compared with a later observation rather than with the same data used to train the model.

3.4 Measures and evaluation metrics

The primary regression metrics are:

- Mean Absolute Error (MAE): measures the average absolute difference between predicted and observed values.

- Root Mean Squared Error (RMSE): penalizes larger errors more strongly through squared differences before taking the square root.
- R-squared (R^2): indicates the proportion of variance in the target that is explained by the model relative to the evaluation data.

The research paper reports approximately $R^2 = 0.84$ and MAE = 0.86 for the hybrid RF–LSTM approach. It also reports baseline R^2 values of approximately 0.68 for RF and 0.25 for LSTM. These values should be treated as the reported study results; the final thesis should insert the exact final experimental statistics if they are available.

Mean Absolute Error measures the average magnitude of prediction error without allowing positive and negative errors to cancel each other. For a set of n observations, MAE is defined as the mean of the absolute differences between observed and predicted values. The research paper reports an approximate hybrid MAE of 0.866. Because the paper does not explicitly state a complete variable-specific unit mapping for this aggregate value, the thesis preserves the reported value and recommends using the final experiment log to state the exact target and unit in the final submission.

RMSE is included in the project's verification methodology and gives greater weight to larger errors because the individual errors are squared before averaging and square-rooting. This is relevant for aviation weather because occasional large forecasting errors can be more operationally significant than many small errors. The supplied research paper does not provide the final RMSE value for the individual RF–LSTM component, so the metric is described and included in the evaluation framework without fabricating a numeric result.

R^2 represents the proportion of variance explained relative to a baseline based on the mean of the observed target. The reported model comparison gives approximately 0.68 for RF, 0.25 for LSTM and 0.84 for the hybrid approach. The hybrid therefore explains substantially more of the target variation in the reported experiment. These figures are presented as approximate because that is the precision provided by the research paper.

The statistical metrics are complemented by aviation transformations. Predicted pressure is converted to Pressure Altitude; predicted temperature is compared with ISA temperature; the resulting ISA deviation is used to calculate Density Altitude. This

does not create a new machine-learning target. Instead, it converts the forecast into a domain-specific quantity that can support interpretation. The system categorizes Density Altitude as Normal, Elevated or Critical using the implemented criteria.

3.5 Data analysis

Data analysis includes descriptive inspection of the meteorological variables, missing-value handling, chronological ordering, feature engineering and model evaluation. The feature engineering described in the paper includes lagged meteorological variables, LSTM windows, sine-cosine encoding of wind direction, dew-point depression, temperature-RH and wind-speed-RH interactions, pressure-wind interaction, RH-squared, cyclic time features, pressure momentum and three- and six-hour rolling averages. These features provide both short-term atmospheric state and temporal context.

The analysis deliberately avoids random shuffling because temporal ordering is part of the prediction problem. The evaluation is therefore intended to reflect a forecasting setting in which the model learns from past observations and predicts later conditions.

The modelling process depends on temporal patterns, so the data analysis includes time ordering, lag creation, rolling averages and cyclic representations of time. These operations allow the model to represent both short-term persistence and periodic effects. The paper mentions hour-of-day and time-of-year representations and three- and six-hour rolling statistics. Such features are particularly relevant to an airport weather series because atmospheric variables can show regular daily patterns while also responding to rapid weather changes.

The central comparative analysis uses the reported RF, LSTM and hybrid results. The hybrid approach achieves the highest reported R^2 among the three. The analysis should not be extended into claims about statistical significance because the supplied paper does not provide repeated-run distributions, confidence intervals or formal significance tests. The correct conclusion is that the reported experiment indicates better aggregate performance for the hybrid model under the tested configuration.

Prediction error should be interpreted in both statistical and aviation terms. A lower MAE means smaller average numerical deviations, while RMSE would expose larger

errors more strongly. For the aviation user, the practical effect also depends on how temperature and pressure errors propagate through Pressure Altitude and Density Altitude. This is why the thesis includes the physical transformation rather than stopping at model metrics.

3.6 Ethics, privacy, security and responsible AI

The TAF states that the study does not involve human participants, personal data or sensitive information and therefore formal human-subject ethical clearance is not required. Nevertheless, the research uses appropriate data-use and attribution practices. Forecast outputs are treated as decision-support information rather than a replacement for official aviation meteorological procedures. Model limitations, uncertainty and validation performance should be communicated clearly when presenting the system.

The TAF states that no human participants, personal data or sensitive information are involved. Nevertheless, responsible research requires correct attribution and appropriate use of meteorological data. The thesis therefore distinguishes source data from derived predictions and documents the data-processing logic. The model should not be represented as a certified replacement for official aviation meteorological services or as an autonomous safety decision-maker.

A forecast model can be accurate on average while still producing occasional large errors. Aviation users should therefore treat model output as decision support and consider the associated verification and confidence information. The system's architecture supports this by retaining forecast timestamps and later comparing predictions with METAR observations. Responsible deployment also requires continued monitoring because atmospheric behaviour can change over time, which is why the project includes model maintenance and retraining mechanisms.

3.7 Reproducibility and research artefacts

Reproducibility requires maintaining the dataset version, preprocessing rules, feature definitions, train/test time boundaries, model configuration, random seeds where applicable, evaluation scripts and result logs. The wider system uses automated data

acquisition and model retraining. The paper describes a FastAPI backend, SQLite/Supabase storage and automated model-maintenance tasks. For the individual component, model artifacts and preprocessing definitions should be versioned so that a reported result can be reproduced.

The methodology can be reproduced conceptually from the documented sequence: obtain historical observations, order and clean them, create lagged and interaction features, construct six-hour sequences, train RF and LSTM models without temporal shuffling, combine their outputs, calculate T+3 predictions and evaluate them against future observations. The supplied paper does not include all code or all hyperparameters. Those artefacts should therefore be attached to the final submission if available rather than inferred.

A useful reproducibility practice for the final thesis is to maintain an evidence ledger containing dataset version, preprocessing version, training period, test period, model configuration, evaluation output and software version. Only the items actually recorded by the project should be populated. The current thesis deliberately identifies undocumented values as “not specified in the supplied source” instead of manufacturing them. This protects academic integrity and makes the final report easier to update when the definitive experiment logs are available.

The research design is quantitative and experimental because it evaluates numerical forecasts against observations recorded for the corresponding target times. The main unit of evaluation is a forecast–observation pair associated with a defined forecast origin and a target three hours later. Each pair should retain timestamps and the variable identity so that errors can be traced back to the exact forecast instance. This record structure is essential for reproducibility: without it, an aggregate metric cannot be checked against the predictions that produced it. The current thesis describes the evaluation approach supported by the project artefacts and does not claim to have reconstructed missing raw prediction logs that were not available in the source material.

Time-series preparation requires particular care because ordinary random splitting can allow observations from the future to influence model development. A

chronological split better reflects the intended forecasting setting: the model is fitted on earlier records and evaluated on later records. Any imputation, scaling, feature selection or transformation that learns parameters from data should be fitted using training data only and then applied to validation and test data. Features calculated from rolling windows must also be aligned so that a forecast at time t does not accidentally include the target observation at t plus three hours. These controls reduce leakage and make the reported error more representative of a genuine future forecast.

Data quality assessment should be performed before model fitting and documented as part of the research artefact. Relevant checks include parsing timestamps consistently, identifying duplicate reports, verifying numerical ranges and units, inspecting missing values, and checking whether time intervals are regular enough for the intended sequence construction. Meteorological observations can be irregular because reports may be delayed, missing or issued at different intervals. A sequence model that assumes regular steps needs an explicit policy for gaps; otherwise, a six-step window may not represent the same elapsed duration for every sample. The project materials describe preprocessing and temporal features, but a complete final report should include the exact gap-handling policy and counts before and after each cleaning stage when those logs are available.

The evaluation metrics answer different questions and should be reported together. Mean Absolute Error expresses the average magnitude of the error in the target's units and is comparatively easy to interpret. Root Mean Squared Error gives greater weight to large errors and can therefore reveal whether occasional large misses are important. R-squared measures the proportion of variance explained relative to a reference mean and can be negative on a test set if forecasts are worse than that reference. None of these metrics alone describes calibration, uncertainty or operational suitability. For that reason, a stronger evaluation would also inspect residual distributions, error by time period and performance during rapidly changing conditions, while preserving the distinction between reported results and proposed future analyses.

The unit of time must remain consistent across data preparation, model fitting and verification. When timestamp fields include a timezone or are converted between local

time and Coordinated Universal Time, the conversion should be performed consistently before temporal features are derived. Otherwise, time-of-day and cyclic features may be shifted, and forecast–observation matching may be incorrect. Duplicate timestamps also require a documented policy because multiple reports can exist for the same nominal time. The policy may retain the most appropriate report or resolve duplicates through a defined rule, but it should not be chosen after looking at test errors. These details are important for a meteorological time series because small timestamp inconsistencies can have disproportionate effects on sequence models.

The analysis should preserve the distinction between model selection and final evaluation. If a validation set is used to choose hyperparameters, the test set should remain untouched until the model configuration is fixed. Repeatedly reviewing test performance and adjusting the model to improve it turns the test set into part of the development process and can lead to optimistic estimates. The available description supports a chronological forecasting approach, but the exact selection protocol should be documented from the original experiment records. A future reproducible version should specify the train, validation and test periods, the criteria for selecting checkpoints and the final configuration used to generate the reported metrics.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional requirements for the individual component are:

Non-functional requirements include reproducibility, temporal integrity, reasonable inference latency, maintainability and clear interpretation of model outputs.

The individual component must accept historical/local meteorological observations, create forecasting features, produce a T+3 temperature and pressure forecast, calculate evaluation metrics when ground truth is available, and expose the results to the wider system. It must also support conversion of predicted temperature and pressure into Pressure Altitude, ISA temperature, ISA deviation and Density Altitude. These requirements are derived directly from the methodology and implementation described in the research paper.

The component should preserve chronological integrity, provide reproducible preprocessing, separate training from inference, and support later verification. Because the complete system is intended for near-real-time operation, the implementation also needs a predictable data interface and a mechanism for model maintenance. The paper reports FastAPI, Supabase, SQLite and scheduled background jobs in the wider solution. Exact response-time or availability targets are not provided in the source and are therefore not invented here.

The design requirements are traceable to the TAF individual row: historical VCBI data collection, time-series preprocessing, multi-variable modelling, evaluation and weather visualization. The final implementation evidence supports the first four strongly. The supplied paper does not provide a detailed temperature heatmap or wind-pattern visualization attributable specifically to this individual component, so such a figure should only be inserted if the actual project artefact exists.

Table 4.1 Functional requirements for the individual component

ID	Requirement
FR1	Ingest historical VCBI meteorological observations.
FR2	Preprocess and chronologically organize time-series data.
FR3	Generate lagged, rolling and interaction features.
FR4	Train an RF regression model.
FR5	Train an LSTM sequence model.
FR6	Combine model outputs into a hybrid forecast.
FR7	Generate three-hour-ahead temperature and pressure predictions.
FR8	Verify predictions against future observations.
FR9	Calculate MAE, RMSE and R^2.

FR10	Calculate aviation-relevant Density Altitude from predicted temperature and pressure.
FR11	Expose outputs for integration with the wider dashboard/system.

4.2 Component architecture within the complete solution

The individual component sits inside the wider hyper-local forecasting and verification architecture. The complete system receives meteorological data, processes the data through multiple specialized prediction modules, stores observations and predictions, verifies forecasts and presents outputs through a dashboard. The individual RF–LSTM module is responsible for temperature and pressure forecasting and for producing inputs to the aviation-calculation layer.

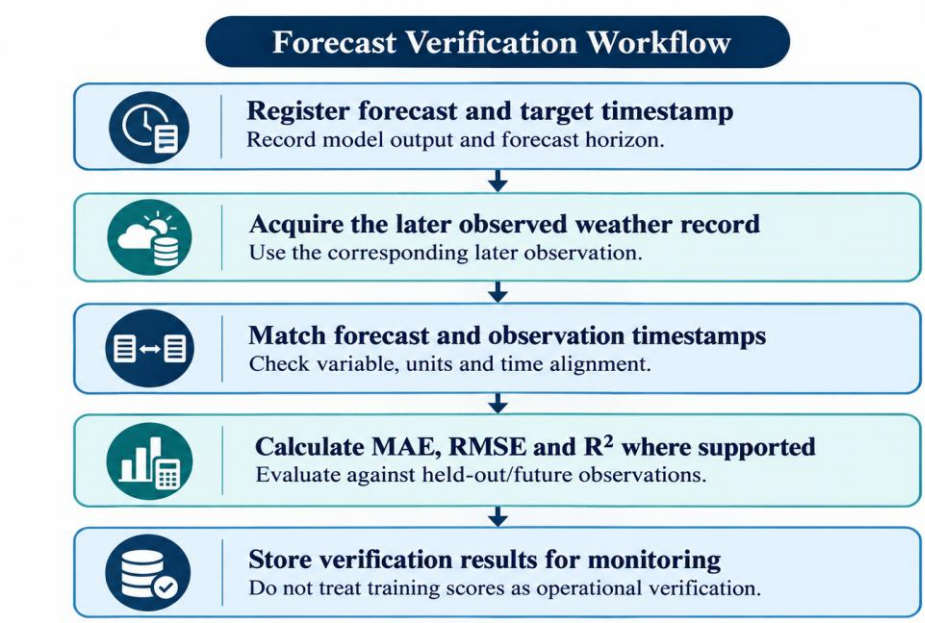


Figure 4.1.Forecast verification workflow

The individual architecture begins with local meteorological observations and proceeds through preprocessing and feature engineering into two complementary

prediction paths. The RF path operates on engineered tabular features. The LSTM path operates on sequential windows. Their outputs are combined into a T+3 forecast. The forecast is then passed to the aviation-calculation stage and the verification store. This architecture makes the individual contribution a self-contained predictive service while still allowing it to participate in the complete group system.

The group paper states that the overall system uses FastAPI for backend services, Supabase for cloud synchronization and a React dashboard for presentation, with SQLite used for local meteorological storage. The individual component's integration boundary is therefore the prediction result and associated metadata. Other modules can consume the pressure forecast while the verification process can later attach the corresponding future observation and error metrics.

4.3 Design alternatives and decisions

The hybrid design is therefore motivated by complementarity rather than by assuming that one algorithm is universally superior. The reported results support the decision because the hybrid approach achieved the strongest reported R^2.

RF and LSTM were retained as complementary alternatives rather than forcing a single algorithm. RF provides a strong tabular representation of engineered relationships, while LSTM provides a sequence representation. The reported results show that neither baseline should be assumed to dominate without testing: RF achieved approximately 68% R^2 while LSTM achieved approximately 25% R^2 in the reported experiment. The hybrid approach was selected because it achieved approximately 84% R^2.

The existence of a lower reported LSTM baseline demonstrates why a deep-learning-only strategy should not be assumed to be superior. With local tabular features, an ensemble tree method can capture nonlinear relationships effectively. The hybrid design therefore uses deep learning where temporal structure is important and a tree ensemble where engineered meteorological interactions are informative. This is a practical rather than ideological model-selection decision.

The system uses a transparent physical transformation for Density Altitude rather than training a separate black-box model to predict Density Altitude directly. This choice

improves interpretability because users can see how predicted pressure and temperature lead to the final aviation metric. It also reduces the need to create another supervised target and keeps the machine-learning component focused on the atmospheric variables for which local observations are available.

Table 2.2 Model alternatives considered

Alternative	Strength	Limitation	Decision
Random Forest only	Strong nonlinear tabular learning	Limited direct sequence modelling	Used as hybrid baseline/component
LSTM only	Captures temporal sequence dependencies	Reported lower baseline R^2 than RF in this study	Used as complementary sequence model
RF–LSTM hybrid	Combines nonlinear feature learning and sequence learning	More complex pipeline	Selected for individual component
Single neural model	Unified architecture	Would not directly exploit RF's tabular ensemble behaviour	Not selected

4.4 Detailed design

The detailed design contains the following logical stages:

- Data ingestion layer: historical observations and current METAR observations.
- Preprocessing layer: chronological ordering, duplicate removal and missing-value treatment.
- Feature engineering layer: lag features, rolling statistics, interactions and cyclic time features.
- RF branch: nonlinear tabular prediction from engineered features.
- LSTM branch: sequential learning from time-windowed observations.
- Fusion layer: combination of RF and LSTM outputs.
- Forecast layer: three-hour temperature and pressure values.
- Aviation layer: PA, ISA temperature, ISA deviation and DA calculation.

- Verification layer: matching forecasts with future observations and calculating errors.
- Integration layer: storing and exposing predictions to the wider application.

For the reported aviation transformation, pressure altitude is calculated from predicted pressure using PA(feet) = (1013.25 − predicted pressure) × 30. ISA temperature is then derived using a standard sea-level temperature of 15°C and a lapse rate of 2°C per 1000 feet. ISA deviation is the difference between predicted temperature and ISA temperature, and Density Altitude is calculated using DA = PA + (118.8 × ISA deviation).

The feature layer contains lagged meteorological variables, temporal encodings, rolling statistics and interaction variables. Wind direction is represented through sine and cosine components to avoid the discontinuity between 359° and 0°. Dew-point depression provides a measure of the separation between dry temperature and dew point. Temperature–RH and wind-speed–RH interactions represent combined atmospheric states. pressure–wind interaction and pressure momentum provide additional representations of recent pressure behaviour.

The model layer consists of an RF regressor and an LSTM sequence model. The RF path receives the engineered tabular representation. The LSTM path uses a six-hour window. The research paper states that their outputs are combined to provide the T+3 temperature and pressure forecast. The exact ensemble weighting is not disclosed in the supplied source; therefore, the final implementation documentation should insert that value only from the actual code or experiment record.

The interpretation layer applies the equations reported in the paper. Pressure Altitude is calculated from standard sea-level pressure and predicted pressure. ISA temperature is calculated from altitude and the stated lapse rate. ISA deviation is the difference between predicted temperature and ISA temperature. Density Altitude is then calculated by adding the temperature-deviation term to Pressure Altitude. The result is categorized using the system's implemented thresholds.

4.5 Standards and professional practice

The project uses aviation weather concepts including METAR and TAF and refers to ICAO-related meteorological requirements in the wider research. The system should be treated as a research decision-support tool. Operational deployment would require additional validation, governance, domain approval and compliance with applicable aviation meteorological procedures.

The project references the FAA Pilot's Handbook and ICAO meteorological service guidance as professional context. The individual component does not claim certification. Instead, it uses aviation terminology and calculations to make the output more meaningful to aviation users. Any operational deployment would require additional validation, governance and acceptance beyond the academic prototype described here.

Professional practice also requires versioned data processing, clear model ownership, separation of training and inference environments, logging of predictions and later verification. The group system's MLOps design supports these principles through scheduled data ingestion and retraining. The thesis treats these practices as engineering controls rather than as evidence that the model is operationally certified.

The design separates data acquisition, preprocessing, prediction, interpretation, persistence and verification because each stage has a different responsibility and failure mode. Acquisition determines which observations enter the workflow; preprocessing converts them into consistent model inputs; the prediction component generates future estimates; the interpretation stage derives aviation-related quantities; and verification compares stored forecasts with later observations. Keeping these responsibilities distinct improves traceability and makes it easier to test one stage without treating the entire system as a single opaque process. It also helps avoid attributing a data-ingestion defect to the learning algorithm when the source of an error lies elsewhere in the pipeline.

The component boundary is important in a group project. The wider system includes other forecasting modules, but this individual thesis evaluates the RF–LSTM temperature and pressure component. Interfaces to other modules are discussed only

to explain how the individual outputs can be consumed by the complete solution. The work does not claim that the author developed the other members' wind, cloud, visibility or separate humidity-related models. A clearly defined boundary improves academic attribution and also supports software engineering practice: a module can be integrated through a documented contract without duplicating or claiming responsibility for its internal implementation.

The forecast record should include, at minimum, a forecast-generation timestamp, target timestamp, model identifier or version, predicted temperature, predicted pressure and the status of any derived calculation. When available, it is useful to retain the input observation window or a reference to it, because this permits later auditing of unusual forecasts. Verification results should be stored separately from the original prediction so that a later observation does not silently alter what the system originally forecast. This design principle preserves an immutable distinction between prediction-time information and outcome-time information, which is especially important when metrics are recalculated after new observations arrive.

A user-facing display should distinguish forecast values from observed values and clearly show units and timestamps. If a Density Altitude value or category is displayed, the calculation method and category definitions should be documented and consistent across the interface and the research evaluation. Missing or stale inputs should lead to an explicit unavailable or warning state rather than a plausible-looking number produced from incomplete data. The interface should also avoid language that implies an AI output is an official forecast or a safety guarantee. These are design requirements for responsible presentation; they do not imply that every proposed safeguard has already been implemented in the prototype.

The design should also account for failure states rather than describing only the successful path. The source API may be unavailable, observations may arrive late, required fields may be missing, or a model artifact may fail to load. Each condition should have a defined response, such as retaining the last verified observation with a clear age indicator, marking the forecast unavailable, or logging an error for later investigation. It is unsafe for a forecasting interface to display an old value without showing that it is stale. These safeguards are design recommendations unless the

implementation artefacts confirm that they have already been built, and they should be identified accordingly in the final system documentation.

The separation between model output and derived output supports modular testing and maintenance. If the prediction algorithm is retrained or replaced, the Density Altitude transformation can remain a separate deterministic function provided its inputs and units remain compatible. Conversely, if the physical calculation is corrected, it can be tested independently without retraining the forecasting model. This modularity reduces the risk that a change in one stage creates an unnoticed defect in another. It also helps make the system easier to explain during a demonstration or viva, because the student can describe which outputs are learned by the model and which are calculated from established equations.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

The exact library versions, operating system and hardware configuration should be inserted from the final development environment before submission. The TAF identifies Python, TensorFlow, PyTorch, Scikit-learn, Pandas and NumPy as planned technologies.

The TAF identifies Python as the primary programming language and lists Scikit-learn, TensorFlow and PyTorch among the intended machine-learning frameworks, together with Pandas and NumPy for data processing. The research paper specifically describes RF, LSTM, PyTorch Forecasting for the TFT wind module, XGBoost, LightGBM, FastAPI, Flask, Supabase, SQLite and React.js across the complete system. This thesis records these technologies as project technologies and distinguishes the individual RF–LSTM component from the other modules.

The research paper states that current METAR observations are stored in SQLite and that the wider system uses Supabase for cloud synchronization and real-time broadcasting. FastAPI provides backend prediction and validation services. These technologies create a separation between data storage, model inference and dashboard presentation. The individual model therefore does not need to own the complete user

interface; it produces forecast information that can be consumed by the system's service layer.

Table 3.1 Technology stack and purpose

Technology	Purpose
Python	Data processing and machine-learning implementation
Scikit-learn	Random Forest and evaluation utilities
TensorFlow / PyTorch ecosystem	Deep-learning / LSTM implementation
Pandas / NumPy	Data processing and numerical operations
SQLite	Local meteorological data and prediction storage described in the paper
Supabase	Central cloud data storage/synchronization described in the paper
FastAPI	Backend prediction, data acquisition and model management services
React.js	Wider dashboard visualization

5.2 Component implementation

The implementation begins with chronological meteorological data preparation. Features are created to represent both instantaneous conditions and recent trends. The RF branch receives engineered tabular features. The LSTM branch receives time-windowed sequences. Their outputs are combined for temperature and pressure prediction. The predicted values are then passed to the aviation calculation layer.

The research paper reports that the temperature and pressure subsystem uses the hybrid RF–LSTM approach and combines the model outputs for a three-hour temperature forecast and pressure forecast. The implementation therefore represents a hybrid machine-learning pipeline rather than an independently trained single predictor.

The implementation orders observations chronologically, removes duplicates, handles missing values and constructs temporal features. Where normalization is required,

Min-Max scaling is applied. The key design principle is that preprocessing used during inference must be consistent with the preprocessing used during training. The source describes the procedure but does not provide the exact code listing in the supplied paper; the final project repository should therefore be treated as the authoritative source for implementation-level details.

The RF implementation uses meteorological and lagged features to model nonlinear relationships. The research paper reports approximately 68% R^2 for the RF baseline. The exact hyperparameters are not given in the paper. Consequently, this thesis does not invent values for tree count, depth, split criteria or feature sampling. Such parameters should be inserted from the actual training configuration if required by the final assessment.

The LSTM implementation uses six-hour windows to capture sequential dependencies. The research paper reports approximately 25% R^2 for the standalone LSTM result. Again, exact architecture details such as number of layers, hidden units, dropout and optimizer settings are not stated in the supplied source. The thesis therefore documents the confirmed sequence-window design and performance while leaving undocumented architecture values to the actual code artefact.

The hybrid model combines the RF and LSTM outputs to generate the final T+3 forecast. The reported R^2 is approximately 84% and MAE approximately 0.866. The improvement over the RF and LSTM baselines is the principal quantitative evidence supporting the individual model design. The result should be reproduced from the final code before final submission if the exact experiment is rerun.

5.3 Integration with the other components

The individual component is integrated with the wider system through the backend and shared data stores. The paper describes FastAPI for real-time data retrieval, inference, model management and retraining, with Supabase and SQLite used to store observations, predictions and verification logs. The React dashboard displays real-time weather, predictions, aviation parameters and warnings. The temperature and pressure outputs contribute to aviation calculations such as Density Altitude and hazard interpretation.

The wider system uses the same meteorological observation stream to support multiple forecasting modules. The individual pressure forecast can therefore be combined with wind, cloud, visibility, dewpoint and RH outputs. This modular design avoids forcing all variables into a single monolithic neural network and allows each subproblem to use an appropriate modelling technique.

The individual forecast is saved with a target timestamp so that a future METAR observation can later be matched. This is the key integration point between prediction and verification. The group paper also describes a confidence and validation concept, although the individual source does not provide a complete confidence-score formula. The thesis therefore describes the timestamp-based verification process as confirmed and treats the exact confidence calculation as a system-level implementation detail requiring repository evidence.

5.4 Verification and software testing

The final submission should attach or reference selected test cases and detailed logs in an appendix. The current research paper provides aggregate results but not a complete software-test case inventory.

Model verification includes comparing predictions with future observations using MAE, RMSE and R^2. The comparison between RF, LSTM and hybrid provides an internal model-selection test. The downstream Density Altitude calculation provides a deterministic transformation that can be checked independently using known inputs. These two testing levels—statistical and deterministic—help isolate whether an error originates in the learned forecast or in the aviation transformation.

A robust implementation should verify chronological ordering, duplicate removal, missing-value handling, feature-window construction and target shifting before model training. These checks are especially important because an indexing error could create data leakage. The source establishes chronological splitting without shuffling, which is the key anti-leakage control documented in the current evidence.

Integration tests should verify that a valid METAR observation can enter the data pipeline, required features can be generated, the model can produce a T+3 forecast, and the result can be stored with its target timestamp. Later, a matching observation

should trigger verification. Exact automated test-case results are not provided in the supplied paper, so this thesis presents the test design and only claims quantitative outcomes that are actually reported.

Table 5.2 .Proposed software test cases and expected outcomes

Test area	Test objective	Expected outcome
Data validation	Check chronological order, duplicates and missing values	Clean time-series data
Feature validation	Check lag and rolling features	Correct temporal alignment
RF model	Evaluate nonlinear prediction	Valid temperature/pressure predictions
LSTM model	Evaluate sequence prediction	Valid sequence-based predictions
Hybrid model	Verify output fusion	Combined forecast generated
Timestamp verification	Match T+3 forecasts to future observations	Correct forecast/observation pairs
Metric calculation	Compute MAE, RMSE and R^2	Correct evaluation values
Aviation calculation	Validate PA, ISA and DA formulas	Consistent aviation indicators

5.5 Deployment and operation

The wider system includes automated METAR acquisition and model-maintenance workflows. The paper describes automated model retraining and real-time verification, including monthly training for the temperature/pressure system, on-demand training through a FastAPI call and hot reloading of updated RF and LSTM models. These mechanisms support continued operation when weather behaviour changes over time.

The group paper reports real-time METAR acquisition through the Aviation Weather Center API, parsing and storage in Supabase, and background jobs for data acquisition and retraining. This creates a continuous path from observation to model input. The individual forecasting component participates in that pipeline without requiring a separate manual data-upload process during normal operation.

The system includes automated or scheduled model maintenance. The paper describes monthly retraining and hot reloading for parts of the system. This is relevant to the individual component because weather relationships can change and a model trained on older data may gradually become less representative. The MLOps approach provides a mechanism for updating the model as additional local observations become available.

Implementation should be described in terms of repeatable stages rather than only a list of technologies. The workflow begins with the source observation records, validates and transforms the data, constructs temporal features, creates model-ready matrices or sequences, fits the component models, combines their outputs, and stores forecasts for later verification. Each stage should have an identifiable input and output so that intermediate artefacts can be inspected when a failure occurs. The current project description supports the use of RF, LSTM, temporal features and a hybrid forecast; exact package versions, hardware specifications and complete training parameters should be added from the actual environment record rather than inferred after the fact.

Feature engineering is a critical implementation step because meteorological conditions have both persistence and cross-variable relationships. Lag features represent prior measurements, rolling statistics summarize recent conditions, and time features can represent recurring daily or annual cycles. Interaction features can expose relationships such as temperature with relative humidity or pressure with wind. However, every engineered feature must be available at the forecast origin. A feature that is calculated from a future observation, or from a centered rolling window, would leak information and inflate test performance. The implementation should therefore be checked not only for whether a feature can be calculated, but also for whether it could legitimately be known at prediction time.

Hybrid integration can be implemented in different ways, including averaging predictions, combining model outputs through a learned meta-model, or using one model's representation as input to another. The exact combination rule materially affects the interpretation of the results. The supplied research material identifies an RF–LSTM hybrid and gives aggregate performance values, but a complete replication

record should explicitly document the fusion rule, training sequence, target alignment and any weighting parameters. These details must be transcribed from the implementation or source paper. They should not be guessed from the word 'hybrid', because different fusion strategies can produce substantially different behaviour even when they use the same component algorithms.

Testing should include data-level, model-level and integration-level checks. Data-level tests verify timestamp ordering, units, missing-value handling and feature-window boundaries. Model-level tests confirm that prediction arrays have the expected shape, contain finite values and align with target timestamps. Integration tests verify that forecast outputs can be passed to the Density Altitude calculation and saved in the format expected by downstream components. Verification tests check that observations are matched to forecasts by target time rather than by row position alone. These tests are practical safeguards against silent errors and provide evidence of software correctness distinct from evidence of predictive accuracy.

Model artefacts should be versioned together with the preprocessing configuration that was used to create their inputs. A saved estimator alone may not be sufficient to reproduce a forecast if feature order, scaling parameters, lag definitions or category encodings are missing. The same applies to the LSTM sequence length and the transformation used to scale its inputs and targets. The final implementation record should therefore preserve the model file, feature list, preprocessing configuration, training period and package versions where available. If these artefacts are not currently retained, that should be stated as a reproducibility limitation rather than implying that the complete training process can be recreated from the model file alone.

Integration testing should verify the semantics of the exchanged values, not only whether a program runs without an exception. Temperature values should retain the expected unit, pressure values should use the unit required by the Density Altitude equation, and timestamps should identify the forecast origin and target time unambiguously. A numeric value can be syntactically valid yet semantically wrong if it is expressed in an unexpected unit. Tests should therefore include known examples with manually checked calculations and boundary cases such as missing pressure,

invalid temperature or a target timestamp that does not follow the forecast origin by the intended interval. Such tests help prevent plausible but incorrect values from passing through the system.

6 RESULTS AND VALIDATION

6.1 Research results

The research paper reports that the hybrid RF–LSTM approach produced an R^2 of approximately 84% and an MAE of approximately 0.86 for the individual temperature and pressure forecasting subsystem. The reported RF baseline had an R^2 of approximately 68%, while the LSTM baseline had an R^2 of approximately 25%. These results indicate that the hybrid model performed better than either baseline according to the reported R^2 values.

The strongest individual quantitative result reported in the research paper is the comparison between RF, LSTM and the hybrid approach. RF achieved approximately 68% R^2, LSTM approximately 25% R^2 and the hybrid approximately 84% R^2. The hybrid MAE is reported as approximately 0.866. These values indicate that the combined representation performed better than either standalone model in the reported experiment. Because the source uses approximate values, the thesis avoids false precision.

The difference between the RF and LSTM results suggests that the engineered tabular representation was particularly useful for the tested dataset, while the sequence model alone did not perform as strongly. The hybrid improvement indicates that the sequential representation still added useful information when combined with the RF output. This interpretation is plausible from the architecture and consistent with the reported metrics, but it should not be described as a causal proof without a feature or component ablation study.

Table 6.1.Reported RF, LSTM and hybrid results

Model	Reported R^2	Reported MAE	Interpretation
Random Forest	≈ 0.68	Not reported in summary	Strong nonlinear tabular baseline
LSTM	≈ 0.25	Not reported in summary	Captures sequence dependency but lower reported R^2
Hybrid RF–LSTM	≈ 0.84	≈ 0.86	Best reported performance

6.2 Product and technical results

The technical output of the component is a three-hour-ahead temperature and pressure forecast that can be consumed by the aviation calculation and verification layers. The predicted temperature and pressure are used to calculate Pressure Altitude, ISA temperature, ISA deviation and Density Altitude. Density Altitude is then categorized in the wider system as Normal, Elevated or Critical according to airport-specific criteria.

The research paper therefore demonstrates an end-to-end relationship between a machine-learning forecast and an aviation-oriented derived measure. This is important because the individual contribution is not limited to numerical prediction; it also translates the predicted atmospheric state into an aviation-relevant quantity.

The individual forecast is converted into Pressure Altitude, ISA temperature, ISA deviation and Density Altitude. This creates an aviation-specific output rather than a raw pressure pair. The paper states that Density Altitude is categorized as Normal, Elevated or Critical according to system-specific operational criteria. This transformation is an important technical result because it demonstrates how the machine-learning output can be connected to a domain-specific decision-support representation.

The complete system integrates the pressure component with other models and a dashboard. The research paper states that the React dashboard displays observation

data, forecasts, aviation data, validation information and warnings. The individual component therefore contributes one part of a larger decision-support interface. The thesis does not claim that all dashboard features were individually implemented by the candidate.

6.3 User or stakeholder validation

The current research paper describes aviation-oriented outputs and guidance, but it does not provide a quantitative user study with a defined participant sample. Therefore, a formal stakeholder validation result should not be claimed from the available evidence. For the final thesis, if stakeholder evaluation was conducted separately, its sample, instrument, procedure and findings should be inserted here.

The supplied research paper does not provide a completed user survey, usability study or quantified stakeholder-validation result for the individual component. The TAF identifies pilots, air traffic controllers, aviation meteorological officers and airport operational managers as stakeholders, but stakeholder identification is not equivalent to completed validation. Therefore, this section reports the absence of such evidence rather than fabricating survey responses or satisfaction scores.

A future validation study could evaluate whether aviation personnel can correctly interpret predicted temperature, pressure and Density Altitude outputs, whether warnings are understandable, and whether the timing of the information fits operational workflows. Such a study would require an approved instrument and documented participants. It should be treated as future work unless actual evidence is available.

6.4 Integration results

The paper reports integration through FastAPI, shared data storage and the React dashboard. The temperature and pressure predictions contribute to the aviation-parameter and warning layer. Forecast verification compares generated predictions with future METAR observations, while the MLOps layer supports recurring data acquisition and model retraining.

The prediction is stored with a target timestamp and later compared with a future METAR observation. This provides a concrete integration between model inference and ground-truth verification. The same broader architecture also connects predictions to the dashboard and MLOps processes. The key individual result is therefore not only a numerical score but a forecast that can move through the system from data acquisition to verification and presentation.

The temperature and pressure predictions are jointly required for the Density Altitude transformation. This creates a dependency between the two target variables: a forecast error in either input can affect the derived aviation metric. The component's coupled interpretation therefore remains relevant even though the implementation uses RF–LSTM rather than a single three-output model.

6.5 Summary of findings

- The hybrid RF–LSTM approach produced the strongest reported R^2 among the three compared approaches.
- Combining nonlinear feature modelling and sequential modelling improved the reported forecasting performance.
- Predicted temperature and pressure can be transformed into aviation-relevant Density Altitude.
- Chronological training/testing and future-observation verification are appropriate for the forecasting problem.
- The available paper supports the technical performance claims but does not provide a complete independent stakeholder validation study.

The available evidence supports four principal findings. First, local VCBI historical data can be structured for short-term machine-learning forecasting. Second, the reported hybrid RF–LSTM approach outperformed the individual RF and LSTM baselines on the reported R^2 measure. Third, predicted temperature and pressure can be transformed into aviation-relevant Density Altitude. Fourth, the forecast can be integrated into a timestamp-based verification process against future METAR observations. These findings are specific to the reported implementation and dataset.

The current source does not establish the exact performance by individual monsoon period, uncertainty intervals, detailed feature importance, statistical significance, long-term operational reliability or stakeholder usability. These are not treated as completed findings. They are identified as areas where the final project repository, experiment logs or future validation study can provide stronger evidence.

The results chapter distinguishes the values reported in the project research paper from analyses that would require additional experiment files. The paper reports an approximate R-squared value of 0.84 and a mean absolute error of about 0.86 for the hybrid model, while the reported R-squared values for Random Forest and LSTM are approximately 0.68 and 0.25, respectively. These figures suggest that the hybrid approach performed better in the reported comparison. They should not be interpreted as exact per-variable metrics unless the source explicitly defines them that way, and they should not be assigned additional decimal places without access to the underlying evaluation output.

Validation requires that every forecast be paired with the correct later observation. The forecast target is three hours after the origin, so a prediction generated at one timestamp must not be evaluated against the most recent observation if that observation occurs earlier than the target. When exact target times are unavailable, the matching tolerance must be stated and justified. A mismatch can create misleading errors even if the prediction itself is reasonable. The final experiment report should therefore document the number of generated forecasts, the number successfully matched, the number excluded, and the reason for each exclusion where such counts can be retrieved from the verification logs.

A single aggregate score can conceal important patterns. For example, a model may perform well during stable conditions but produce larger errors when pressure or temperature changes quickly. Performance can also differ across time of day, observation density or periods with missing data. A fuller validation should therefore include time-series plots of predicted and observed values, residual plots, histograms of absolute error, and error summaries for defined time intervals. These analyses are recommended extensions unless the corresponding charts and logs are already present

in the implementation. The thesis does not treat them as completed experiments when the supplied material does not demonstrate that they were performed.

The Density Altitude output needs its own validation pathway. First, predicted pressure and temperature should be transformed using the documented equations and units. Second, the same calculation should be applied to the corresponding observed pressure and temperature. Third, the derived forecast should be compared with the derived value from the observations. This allows error propagation to be assessed directly. If the system classifies values into categories such as Normal, Elevated or Critical, the classification thresholds must be documented and tested at their boundaries. The model's numerical accuracy and the category agreement rate are different measures and should be reported separately.

The reported aggregate performance values should be presented with clear definitions of their scope. R-squared is unitless, whereas MAE is expressed in the unit of the target variable. If a source reports one aggregate MAE across multiple targets, the thesis must explain how the targets were combined before that value can be interpreted; if it reports MAE for one variable, the variable must be named. The available source provides approximate figures but does not support every possible disaggregation. Consequently, the thesis preserves the reported level of detail and identifies the need for separate temperature and pressure tables as a future reporting improvement. This avoids giving readers the impression that a single aggregate score fully characterizes both outputs.

A useful validation report should include both numerical summaries and examples of individual forecast behaviour. An aggregate metric provides a compact comparison, while a time-series plot can reveal lagging predictions, systematic bias or errors concentrated around abrupt changes. A residual plot can indicate whether the model consistently overpredicts or underpredicts within certain value ranges. These visual checks are complementary to the formal metrics, not substitutes for them. They should be generated from the held-out forecasts and corresponding observations, with axes labelled in the correct units and the test interval stated. If such plots are added later, the underlying data and generation code should be retained so that the figure can be regenerated for review.

7 DISCUSSION

7.1 Interpretation of findings

The reported performance suggests that the pressure problem contains both nonlinear feature relationships and temporal dependencies. RF can learn nonlinear relationships in engineered tabular features, while LSTM can model sequential dependencies. The improvement to approximately 0.84 R^2 for the hybrid approach is therefore consistent with the intended complementary design. However, performance metrics should be interpreted together with the exact test period, variable-specific errors and distribution of weather conditions.

The aviation transformation also increases the practical relevance of the forecast. A temperature or pressure error has a physical implication for aircraft performance when translated into Density Altitude. Therefore, evaluating only generic ML metrics is insufficient for an aviation decision-support application; the derived aviation quantity should also be examined.

The hybrid result is meaningful because it addresses a modelling limitation rather than simply increasing model complexity. RF and LSTM receive different representations of the same meteorological problem. RF benefits from engineered nonlinear relationships, while LSTM captures sequence structure. The reported improvement from approximately 68% RF R^2 and 25% LSTM R^2 to approximately 84% hybrid R^2 is consistent with complementary information being combined. However, the improvement should be understood within the tested dataset and configuration.

An airport forecast is useful when it supports a decision or provides a clearer representation of risk. The Density Altitude transformation is therefore important to the interpretation. Rather than presenting an unexplained model score, the system connects predicted pressure and temperature to Pressure Altitude, ISA deviation and Density Altitude. This creates a domain-oriented output while retaining the underlying forecast values for verification.

7.2 Comparison with literature baselines and existing solutions

The research differs from general Sri Lankan weather forecasting studies because it is explicitly hyper-local and aviation-oriented. It also differs from global numerical weather prediction because it uses local airport observations and short forecast horizons. Compared with using RF or LSTM alone, the reported hybrid result provides evidence that a combined architecture can better represent the characteristics of the local time series in this experiment.

The approach is also complementary to automated TAF research. Rather than attempting to replace official TAF production, the individual component produces an additional local forecast that can be verified against aviation observations and used in the wider decision-support system.

The reviewed Sri Lankan studies demonstrate that machine learning can be used for weather and monsoon prediction, but the project focuses on a different spatial and operational scale. The individual component uses airport-level observations and a three-hour horizon, with later verification against aviation observations. The contribution is therefore better characterized as contextual specialization than as a direct replacement for earlier general forecasting research.

AI-assisted TAF and aviation-weather research shows that machine learning can support aviation forecasting. The individual component differs by emphasizing hyper-local VCBI observations and a transparent pressure-to-Density-Altitude transformation. The project therefore combines predictive modelling with aviation interpretation and verification rather than generating only a textual or categorical forecast.

7.3 Research contribution

- A hyper-local hybrid RF–LSTM approach for short-term temperature and pressure prediction at VCBI.
- Integration of meteorological feature engineering with both nonlinear tabular and sequential learning.
- A reported performance improvement of the hybrid model over the individual RF and LSTM baselines.

- Transformation of predicted temperature and pressure into Density Altitude for aviation interpretation.
- Integration of forecast generation with future-observation verification.

The methodological contribution is the application of a hybrid RF–LSTM approach to short-term hyper-local aviation weather prediction at VCBI. The feature engineering combines lagged variables, rolling statistics, cyclic time features and meteorological interactions. The model comparison provides evidence for selecting the hybrid approach over the individual baselines in the reported experiment.

The domain contribution is the connection of temperature and pressure prediction to Density Altitude and aviation-oriented verification. This creates a path from local weather observations to an interpretable aviation metric. The result is particularly relevant to the project's stated focus on tropical monsoon environments, although further multi-season and multi-airport validation is still required.

7.4 Product contribution

The product contribution is a forecasting component that supplies temperature and pressure predictions to the wider aviation weather system. These outputs support aviation calculations, dashboard visualization and forecast verification. The component also fits within the MLOps architecture described in the paper, allowing future observations to be used for continued model maintenance.

The individual component contributes a forecast service that can be consumed by the broader aviation weather dashboard. Its outputs include predicted temperature and pressure and derived aviation metrics. The system architecture also supports later verification and model maintenance. This makes the component useful as a decision-support module rather than as a standalone academic model with no operational pathway.

The wider MLOps design provides a mechanism for periodic retraining and continuous ingestion. For a weather model, maintainability matters because concept drift can occur when atmospheric behaviour changes or the data distribution evolves. The individual component benefits from this architecture because its performance can be re-evaluated as new observations become available.

7.5 Limitations and threats to validity

The paper provides the broad dataset description but not the complete distribution of missing values, outliers, station changes or per-period sample counts. Therefore, the thesis cannot claim that the dataset is perfectly homogeneous. The 2019–2026 period is sufficiently broad for the project context, but further analysis should examine whether model performance varies across monsoon regimes and unusual weather events.

The reported results are aggregate values. The paper does not provide confidence intervals, repeated-run variability, a full ablation study or a complete per-season error table for the individual component. These limitations reduce the strength of generalization claims. The most defensible conclusion is that the hybrid performed better in the reported experiment, not that it will always outperform alternative models.

The system is an academic research prototype and should not be represented as a certified aviation forecasting service. Official aviation forecasts and observations remain important sources of information. Any operational deployment would require extensive validation, safety assurance, monitoring, governance and regulatory consideration.

Table 7.1Threats to validity and mitigation

Threat / limitation	Potential effect	Mitigation / future action
Reported summary metrics are approximate	Limits exact comparison	Insert exact final experiment logs and confidence intervals where available
Single airport focus	Limits geographic generalization	Validate at additional Sri Lankan airports
Monsoon and rapid-change conditions	Extreme conditions may be underrepresented	Evaluate separately by season and weather regime
Potential concept drift	Model performance may decline as weather behaviour changes	Use MLOps retraining and drift monitoring
No formal stakeholder study in current paper	Usability claims are limited	Conduct structured aviation stakeholder evaluation
Hybrid architecture is more complex	More maintenance and compute requirements	Version models and automate deployment
Derived DA depends on forecast errors	Atmospheric forecast error propagates into aviation metric	Evaluate DA error and operational thresholds explicitly

7.6 Implications

The findings imply that hyper-local aviation forecasting can benefit from hybrid architectures that combine different learning mechanisms. For Sri Lankan airport environments, local historical observations provide information that can be used to complement larger-scale forecasts. The results also indicate that aviation-oriented ML systems should connect prediction metrics with physically meaningful operational indicators rather than presenting raw model outputs alone.

The results suggest that hybrid modelling is worth investigating for airport-level weather forecasting because local meteorological variables contain both nonlinear relationships and temporal dependencies. Future studies can build on this by performing controlled ablation experiments, uncertainty estimation and regime-specific evaluation.

From an engineering perspective, the component demonstrates the value of separating prediction, verification and physical interpretation. This separation allows model performance to be monitored independently from the deterministic aviation calculations. It also makes it easier to replace or retrain the forecasting model without redesigning the entire dashboard.

The reported comparison is consistent with the hypothesis that combining tabular nonlinear learning and sequential learning can be beneficial for this forecasting task. Random Forest can represent nonlinear interactions in engineered weather features, while LSTM can use the order of observations to learn temporal patterns. A hybrid may draw on both capabilities, but the aggregate comparison does not by itself prove which component contributed most to the improvement. Without an ablation experiment or a documented fusion analysis, the most defensible interpretation is that the combined approach achieved the best reported result under the conditions described by the source paper, not that each feature or architecture choice has been independently validated.

The result must also be interpreted in relation to the evaluation design. A model can appear stronger if it is tested on an easier period, uses a different set of records, or benefits from preprocessing that exposes future information. The thesis therefore treats chronological splitting, target alignment and leakage control as conditions for trusting the comparison. Since the available material does not contain a complete reproducibility package with every prediction and configuration, the performance values are evidence of the reported experiment rather than a fully independent replication. Future work should publish the exact split dates, sample counts, metric definitions and model configuration needed to reproduce the result.

The practical contribution is the connection between forecast generation and aviation-oriented interpretation. Density Altitude provides a useful example of converting weather variables into a quantity that has a clearer operational meaning than an isolated regression score. Nevertheless, the transformation should be treated as a decision-support aid and not as a direct measure of aircraft performance in every operating context. Aircraft type, runway characteristics, loading, wind and other conditions may also affect operational decisions. The individual model therefore

contributes one piece of information to a wider aviation-weather picture and should not be presented as a complete aircraft performance assessment or as a replacement for professional judgement.

The limitations define the strength of the claims that can be made. The research is geographically focused on VCBI, the quantitative results are reported at aggregate precision, and the available source material does not show full uncertainty estimates, a complete ablation study or independent operational validation. These limitations do not remove the value of the prototype, but they restrict generalization. The most appropriate conclusion is that the reported hybrid method is promising for the defined local forecasting task and merits more systematic testing. Broader claims require testing on additional periods, additional airports and a wider range of atmospheric conditions with consistent verification rules.

The findings have implications for how future experiments should be prioritized. Improving the reproducibility of the existing comparison may be more informative than immediately adding a more complex architecture. Once the baseline results are reproducible, targeted ablation experiments can isolate the value of lagged features, rolling averages, interaction features and sequence length. Error analysis can then identify whether the remaining problem is general bias, occasional extreme errors or performance under rapidly changing conditions. This staged approach makes research decisions evidence-driven and reduces the chance of attributing an improvement to a change that was not actually responsible for it.

The work also highlights the importance of communicating uncertainty and limitations to end users. A point forecast does not indicate how confident the model is, and a category derived from that point estimate may hide uncertainty near a threshold. Future versions could investigate prediction intervals or empirically calibrated error bands, provided that these are evaluated on data that was not used to fit the model. Until such information is available, the interface should avoid overstating certainty and should clearly identify the forecast as an estimate. This is particularly important in aviation-related settings, where a technically accurate prototype must not be mistaken for an approved operational service.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This individual research developed and evaluated a hybrid RF–LSTM approach for short-term temperature and pressure prediction within a hyper-local aviation weather forecasting system for VCBI. The approach addresses the individual research problem by combining nonlinear feature modelling with sequential temporal learning. The reported results indicate that the hybrid approach achieved approximately 0.84 R^2 and 0.86 MAE, outperforming the reported RF and LSTM baselines in R^2. The forecasts were also connected to aviation physics through Pressure Altitude, ISA temperature, ISA deviation and Density Altitude calculations.

The individual study demonstrates an evidence-supported approach for short-term hyper-local temperature and pressure forecasting at VCBI. The work begins with airport-level historical observations, applies time-series preprocessing and meteorological feature engineering, develops RF and LSTM models, combines their outputs and evaluates the resulting forecast. The reported hybrid R^2 of approximately 84% is higher than the approximately 68% RF and approximately 25% LSTM baselines. The predicted variables are then transformed into aviation-relevant Density Altitude information and integrated into the wider verification architecture.

The conclusion is deliberately limited to the evidence available in the current research paper and TAF. It does not claim complete operational certification, universal superiority of RF–LSTM, or statistically significant improvement across all monsoon periods. Further evidence is needed for those stronger claims. This boundary is an essential part of an academically responsible conclusion.

8.2 Achievement of Objectives

The objective of collecting and preparing historical VCBI observations is supported by the dataset description. Feature engineering is supported by the detailed list of lagged, rolling, cyclic and interaction features. RF and LSTM model development is supported by the reported architecture. Hybrid T+3 forecasting and evaluation are supported by the reported results. Aviation transformation is supported by the Density

Altitude formulas. Forecast verification is supported by the timestamp-matching procedure, although a complete final verification table is not included in the supplied paper.

Table 8.1. Objective achievement based on available evidence

Objective	Achievement based on available evidence
Collect and prepare VCBI weather data	Achieved; paper reports 2019–2026 historical data and preprocessing.
Engineer temporal and meteorological features	Achieved; multiple lag, rolling, cyclic and interaction features are reported.
Develop RF model	Achieved and used as a baseline/hybrid branch.
Develop LSTM model	Achieved and used as a sequence-learning branch.
Develop hybrid RF–LSTM model	Achieved; combined outputs reported for T+3 temperature and pressure.
Evaluate and verify forecasts	Achieved in the reported methodology using MAE, RMSE, R^2 and future observations.
Translate predictions to aviation information	Achieved through PA, ISA and Density Altitude calculations.

8.3 Individual Research and Product Contributions

The principal individual research contribution is the hybrid RF–LSTM forecasting approach for temperature and pressure and its evaluation within the hyper-local VCBI context. The principal product contribution is the integrated forecasting component that supplies aviation-relevant atmospheric predictions and Density Altitude information to the wider system.

The individual research contribution is the design and evaluation of the hybrid RF–LSTM pressure forecasting subsystem and its connection to aviation interpretation and verification. The product contribution is a reusable forecasting component that can be integrated into the wider VCBI hyper-local weather system. The work is distinct from the wind, cloud/visibility and separate dewpoint/RH components implemented by other group members.

8.4 Future Directions

- Evaluate performance across distinct seasonal and rapidly changing weather regimes, without restricting the model to a single season.
- Report exact variable-level MAE, RMSE and R^2 values with confidence intervals and prediction-error distributions.
- Evaluate the accuracy of derived Density Altitude rather than only the input temperature and pressure forecasts.
- Investigate additional hybrid architectures and hyperparameter optimization while maintaining temporal integrity.
- Expand validation to other Sri Lankan airports to test generalizability.
- Incorporate additional remote-sensing or meteorological observations where appropriate.
- Implement explicit concept-drift monitoring and retraining triggers.
- Conduct structured evaluation with aviation meteorological experts and intended users before operational deployment.

The research paper identifies several future directions: larger datasets, better performance under rapidly changing weather, extended monsoon validation, additional meteorological and remote-sensing information, and implementation at other Sri Lankan airports. For the individual component, particularly valuable extensions include monsoon-specific error analysis, feature ablation, uncertainty estimation, ensemble calibration and comparison with additional temporal models.

The product can be extended with richer verification dashboards, forecast-confidence histories, event-based performance analysis and automated model-health monitoring. Any future aviation-facing feature should preserve the principle that AI forecasts are decision-support information and should remain transparent about uncertainty and validation status.

The study addresses a defined research problem: producing short-horizon temperature and pressure forecasts from airport-level observations and evaluating them in a way that preserves temporal integrity. The implemented workflow combines meteorological feature preparation, Random Forest modelling, LSTM sequence

learning and a hybrid prediction stage. It also connects predicted values to an aviation-related Density Altitude calculation and provides a basis for comparison with later observations. The work is therefore both a modelling exercise and a systems exercise, because forecast validity depends on the data pipeline, timestamp handling and downstream calculations as well as on the learning algorithms.

The objectives have different levels of evidence. The available project paper documents the general dataset context, feature families, model approach and aggregate model comparison. The three-hour forecasting horizon and Density Altitude transformation are also described. However, exact variable-specific scores, uncertainty intervals, detailed ablation results and full reproducibility parameters are not available in the same level of detail. The thesis therefore distinguishes implemented elements from analyses that remain necessary for a stronger evaluation. This distinction is academically important because an objective should be marked as fully achieved only when its stated deliverable and corresponding evidence can be demonstrated.

The most valuable next step is to strengthen the evaluation record before adding complexity to the model. This includes preserving the prediction file, documenting the exact chronological split, reporting temperature and pressure metrics separately, and calculating confidence intervals or repeated-period variability where appropriate. A feature ablation study can then test whether lag variables, rolling summaries and meteorological interactions contribute measurable improvement. These steps would explain not only whether the hybrid performs well, but why it performs well and under what conditions its performance degrades. Such evidence would make the research easier to reproduce and more useful to future developers.

Further development should preserve the boundary between experimental research and operational aviation service. Before any operational use, the component would need extensive prospective verification, monitoring of data quality and model drift, explicit failure handling, and review by qualified meteorological and aviation stakeholders. The interface should communicate uncertainty and clearly identify the source and time of every forecast. Expansion to other airports should occur only after the model has been evaluated against local data and conditions at those locations.

These future directions build on the current prototype without overstating the readiness or generality of the reported results.

The contribution of the individual study is best understood as a complete experimental path rather than a claim about one algorithm alone. It includes defining the forecast target, preparing historical observations, constructing temporal and meteorological features, comparing RF and LSTM with a hybrid approach, translating the forecast into an aviation-related derived quantity and considering how the prediction can be verified later. Each stage contributes to the reliability of the final result. This end-to-end perspective is useful for future work because improvements can be made at the stage where evidence identifies a weakness, rather than assuming that every error must be solved by changing the model architecture.

For final submission, the strongest remaining improvements are evidence-focused: insert exact model settings from the implementation, include the final split dates and sample counts, provide separate target-level metrics if available, and ensure that every table and figure has a traceable source. The reference list should be checked against the citations in the text, and any placeholder information on the cover page should be completed. These are not cosmetic details; they determine whether a reader can follow the argument and assess the reliability of the reported findings. The thesis should be submitted only after these factual checks are completed against the original project artefacts.

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GLOSSARY

Term	Definition

Hyper-local forecasting	Forecasting focused on a specific local site such as an airport rather than a broad regional domain.
METAR	Routine aviation meteorological observation/report describing current conditions.
TAF	Terminal Aerodrome Forecast for an airport.
pressure	Aviation pressure/altimeter-setting quantity used to relate atmospheric pressure to altitude calculations.
Density Altitude	An aviation-relevant altitude measure influenced by pressure and temperature.
Random Forest	An ensemble machine-learning method based on multiple decision trees.
LSTM	Long Short-Term Memory recurrent neural-network architecture for sequential data.
Hybrid model	A model architecture that combines outputs or capabilities of multiple modelling approaches.
Concept drift	A change over time in the relationship between inputs and the target that can reduce model performance.

APPENDICES

Appendix A: Evidence Status and Source Traceability

This appendix records what is directly supported by the supplied Topic Assessment Form (TAF) and research paper and what should be completed from the final repository or experiment logs. The purpose is to prevent unsupported numbers from entering the thesis.

The research paper reports more than 120,000 historical VCBI samples for 2019–2026; RF R^2 of approximately 68%; LSTM R^2 of approximately 25%; hybrid RF–LSTM R^2 of approximately 84%; and hybrid MAE of about 0.866. It also reports separate cloud/visibility and dewpoint/pressure results in the group paper. Only the pressure-related results are treated as individual quantitative evidence in the main analysis.

Exact RF hyperparameters, exact LSTM layer sizes, exact hybrid weighting, exact RMSE for the individual component, exact train/test row counts, confidence intervals, per-monsoon performance tables, user-survey results, and final response-time measurements are not stated in the supplied source. These details should be added only if they can be verified from the actual project artefacts.

Appendix B:Individual Component Data Dictionary

The source describes temperature, wind speed, wind direction, pressure, humidity and dew point as important meteorological variables. Temperature and pressure are the principal outputs of the individual RF–LSTM subsystem. Other variables are used as input features where available.

The feature-engineering description includes hour-of-day and time-of-year representations using sine and cosine encodings. Lagged observations and three-hour and six-hour rolling averages represent recent temporal context.

Dew-point depression, temperature–RH interaction, wind-speed–RH interaction, pressure–wind interaction, RH squared and pressure momentum are derived features described in the research paper. Wind direction is encoded using sine and cosine to avoid circular discontinuity between 0° and 360°.

Appendix C: Preprocessing and Feature-Engineering Catalogue

Observations are ordered chronologically before modelling. This is essential for a forward-forecasting experiment because the target at T+3 must occur after the information available at time T.

The research paper states that duplicate records are removed and missing values are processed. The exact imputation strategy is not specified in the supplied paper. Therefore, the final report should include the actual preprocessing method from the repository if required.

Min-Max normalization is applied where required. The thesis does not assume that every Random Forest feature was normalized because tree-based models generally do not require feature scaling. It records only that normalization was applied where required.

The LSTM uses six-hour windows. Each sequence therefore represents a continuous temporal context rather than an isolated observation. The exact sampling frequency and tensor dimensions should be confirmed from the final implementation.

Sine-cosine encodings allow periodic time variables to be represented without a false numerical boundary. For example, the final hour of a daily cycle and the first hour of the next cycle remain mathematically close.

Interaction variables represent combined atmospheric conditions. They do not establish physical causation. Instead, they provide structured input features from which the model can learn nonlinear associations.

Appendix D:Mathematical Specification

Mean Absolute Error (MAE) $MAE = (1/n) \Sigma |y_i - \hat{y}_i|$

MAE summarizes the average absolute difference between observed and predicted values.

Root Mean Squared Error (RMSE)

$RMSE = \sqrt{[(1/n)\ \Sigma\ (y_i - \hat{y}_i)^2]}$ RMSE gives larger errors greater influence because the differences are squared before averaging.

Coefficient of Determination (R^2)

$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$ R^2 measures model performance relative to a mean-based reference.

Pressure Altitude (PA) -PA (feet) = (1013.25 − predicted pressure) × 30 This is the transformation reported in the research paper. The pressure unit used in the implementation must be verified before final submission.

ISA Temperature -ISA Temperature (°C) = 15 − [2 × (PA / 1000)] This is the simplified lapse-rate relationship reported as being used by the system.

ISA Deviation -ISA Deviation (°C) = Predicted Temperature − ISA Temperature This represents the difference between predicted temperature and ISA temperature.

Density Altitude (DA) - DA (feet) = PA + (118.8 × ISA Deviation)

The system categorizes the calculated Density Altitude as Normal, Elevated or Critical according to its implemented criteria.

Relative Humidity Relationship –

$RH = 100 \times \exp[(17.67Td/(Td + 243.5)) - (17.67T/(T + 243.5))]$

The August–Roche–Magnus relationship is used primarily by the separate dewpoint/relative-humidity subsystem. It demonstrates the mathematical relationship among temperature, dew point and relative humidity.

In these equations, y_i represents an observed value, $\hat{y}_i$ represents a predicted value, $\bar{y}$ represents the mean observed value, n represents the number of evaluated samples, T represents temperature, and Td represents dew-point temperature.

Appendix E: Model Comparison Evidence Table

The current research paper reports the following approximate model performance values. Random Forest (RF): R^2 of approximately 68%. Long Short-Term Memory (LSTM): R^2 of approximately 25%. Hybrid RF–LSTM: R^2 of approximately 84% and MAE of approximately 0.866.

These values should be treated as approximate and verified against the final experiment logs. The comparison supports the hybrid approach in the reported experiment. However, it does not establish statistical significance, universal superiority, or performance across every seasonal weather condition.

Appendix F: Forecast Verification Protocol

Every forecast is associated with a target timestamp to preserve the temporal identity of the prediction.A future METAR observation becomes available through the data-acquisition pipeline.The verification stage identifies the future observation corresponding to the forecast target timestamp.The predicted and observed values are compared using MAE, RMSE and R^2 for the regression targets. Forecast and verification information can be retained so that model performance can be monitored over time.The wider MLOps architecture uses updated observations for periodic retraining and continued validation.

Appendix G:Software Test Catalogue

Test 1: Chronological Ordering Input: Observations provided in an unordered sequence. Expected result: Observations are sorted chronologically before feature construction.Evidence required: Execution results from the final test log.

Test 2: Duplicate Record Handling Input: Duplicate timestamp or observation records. Expected result: Duplicate records are removed according to the implemented preprocessing rule.Evidence required: Preprocessing test results or logs.

Test 3: Missing Meteorological Values Input: Records containing missing meteorological values. Expected result: Records are processed according to the actual project preprocessing implementation. Evidence required: Confirmation of the missing-value handling strategy.

Test 4: Lag Feature Construction Input: Chronologically ordered time-series observations. Expected result: Lagged features refer only to earlier observations and do not use future information. Evidence required: Feature-generation test results.

Test 5: LSTM Sequence Preparation Input: A valid time series. Expected result: The LSTM receives the defined six-hour historical window. Evidence required: Confirmation of the sampling frequency and tensor dimensions.

Test 6: Three-Hour Forecast Target Input: An observation recorded at time T. Expected result: The prediction target corresponds to the observation three hours ahead. Evidence required: Verification of target timestamp alignment.

Test 7: Random Forest Prediction Input: A valid engineered feature row.Expected result: The Random Forest model returns a numerical forecast.Evidence required: Model test output.

Test 8: LSTM Prediction Input: A valid six-hour sequence. Expected result: The LSTM model returns a numerical forecast. Evidence required: Model test output.

Test 9: Hybrid Forecast Generation Input: Random Forest and LSTM outputs. Expected result: The hybrid forecasting stage returns a combined forecast using the configured method. Evidence required: Confirmation of the combination method in the source code.

Test 10: Aviation Parameter Calculation Input: Predicted pressure and temperature.Expected result: Pressure Altitude, ISA Temperature, ISA Deviation and Density Altitude values are calculated. Evidence required: Verification of the equations and units against the implementation.

Test 11: Forecast VerificationInput: A stored forecast and a later METAR observation.Expected result: The correct target observation is matched using its timestamp.Evidence required: Timestamp-matching test results.

Test 12: Evaluation Metrics Input: Prediction and observation pairs.Expected result: MAE, RMSE and R^2 are calculated according to their mathematical definitions.Evidence required: Comparison with final metric logs.

Test 13: Incomplete Observation HandlingInput: An incomplete meteorological observation.Expected result: The service rejects or safely handles the record according to the actual implementation. Evidence required: Confirmation of the implemented error-handling behaviour.

Test 14: Model Retraining Input: Newly available training data.Expected result: The retraining workflow updates the model according to the configured schedule. Evidence required: Scheduler and training logs.

Appendix H:Requirements Traceability Matrix

Requirement R1: Historical VCBI Data CollectionSource: TAF individual task. Implementation: Data acquisition and preprocessing. Status: Supported by the supplied project description.

Requirement R2: Time-Series Preprocessing Source: TAF individual task. Implementation: Chronological ordering, deduplication, missing-value handling and temporal feature construction. Status: Supported, subject to confirmation of the exact preprocessing implementation.

Requirement R3: Coupled Meteorological Modelling Source: TAF novelty statement.Implementation: Random Forest and LSTM use interacting meteorological features. Status: Supported with implementation refinement.

Requirement R4: Three-Hour-Ahead Forecasting Source: Research paper methodology. Implementation: Hybrid RF–LSTM forecasting output.Status: Supported by the supplied research description.

Requirement R5: Forecast Evaluation Source: TAF and research paper. Implementation: MAE, RMSE and R^2. Status: Evaluation methods are supported; complete final statistics remain to be confirmed.

Requirement R6: Aviation Interpretation Source: Research paper methodology. Implementation: Pressure Altitude, ISA Temperature, ISA Deviation and Density

Altitude calculations. Status: Supported by the supplied research description.

Requirement R7: Future METAR Verification Source: Research paper methodology.Implementation: Matching forecasts with future observations using timestamps.Status: Supported by the supplied research description. Requirement R8: Stakeholder Validation Source: Thesis template. Implementation: No completed user study is evidenced in the supplied source. Status: Not yet evidenced.

Appendix I: Feature-to-Purpose Matrix

Lagged observations represent recent persistence and short-term atmospheric conditions.LSTM sequence windows represent sequential temporal dependencies.Wind direction sine-cosine features represent circular direction without a discontinuity between 0° and 360°.Dew-point depression represents the difference between temperature and dew point.Temperature–RH interaction represents combined thermal and moisture conditions.Wind-speed–RH interaction represents combined wind and moisture conditions.Pressure–wind interaction represents an interaction between pressure and wind.RH squared allows nonlinear relative-humidity effects to be represented.Daily and seasonal sine-cosine features represent periodic time patterns.Pressure momentum represents recent pressure movement. Three-hour rolling averages summarize and smooth recent atmospheric conditions. Six-hour rolling averages provide a longer temporal summary of recent conditions.